New Sites, New Services: Evidence from a Randomized Dialysis Payment Reform

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Abstract

I use the geographic randomization of Medicare's End-Stage Renal Disease Treatment Choices (ETC) Model to study how payment incentives shape the services that new facilities supply after entry. Linking facility entry histories to annual cost reports, I follow fixed entry cohorts and incumbents through 2024 and compare their contributions with covered-market supply. Statistically significant increases in home dialysis–providing facilities and Medicare home treatments are concentrated among early entrants. The first payment-period cohort contributes an additional 0.125 providing facilities and 183.01 Medicare home-treatment equivalents per hospital referral region on average, with positive treatment responses continuing through follow-up. A similar pattern is not evident among incumbents or across the covered market. For the first payment-period cohort, an accounting decomposition assigns most of the supply increase to entrant numbers, with a smaller contribution from service per entrant. Higher reported training and administrative staffing accompany this expansion. Supplementary market estimates show lower long-term catheter use and higher fistula use; effects on mortality, hospitalization, and geographic access remain imprecisely estimated. These findings suggest that new facilities can be a margin of recorded supply adjustment to payment reform, while net market expansion and broader welfare gains remain uncertain.

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1 Introduction

Payment reform can change the supply of care through adjustments at existing facilities and the services supplied by new entrants. New facilities choose their staffing, training arrangements, and service offerings as they establish operations,

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while incumbents adapt an existing organization. These initial choices may shape supply over several years, even when entrants represent a small share of the market. Understanding this response requires following the care supplied after entry and placing it alongside changes among established providers. I study these adjustments in home dialysis, asking how a randomized payment reform affects the services supplied by successive entry cohorts and how their contributions relate to incumbent and market-wide supply.

Medicare's End-Stage Renal Disease Treatment Choices (ETC) Model provides the randomized comparison. Beginning in January 2021, the model changed payments to encourage home dialysis and transplantation. Within the randomized study frame, 91 of 302 hospital referral regions (HRRs) were assigned to treatment and 211 to control, with assignment stratified across four Census regions (Ji et al. 2022). Home dialysis requires patient and caregiver training, supplies, and continuing clinical support, making the organization of these services relevant to providers' responses. ETC combined a temporary home-dialysis payment bonus with continuing performance incentives (Centers for Medicare & Medicaid Services 2020a). This setting allows me to follow the supply contributions of new facilities as they age and the payment schedule changes.

The analysis relates to evaluations of ETC and to the broader literature on how reimbursement shapes provider behavior. ETC evaluations study home dialysis use, transplantation, and related patient outcomes, with findings that vary across populations and follow-up periods (Centers for Medicare & Medicaid Services 2025c; Dixit et al. 2026; Ji et al. 2022; Johansen et al. 2023; Koukounas et al. 2024). More broadly, studies of physician and hospital payment changes document adjustments in treatment supply, inputs, and technology adoption (Acemoglu and Finkelstein 2008; Clemens and Gottlieb 2014). In dialysis markets, Dai (2014) examines service choice under regulated prices. Following newly established locations connects these questions about reimbursement and service provision to the ETC setting, showing how facilities entering under the reform's payment incentives contribute to subsequent home dialysis supply.

I link annual dialysis facility cost reports from the Healthcare Cost Report Information System (HCRIS) to facility histories from the Provider of Services (POS) files. Cost reports measure treatment volume, training, staffing, and expenditure; certification histories, historical addresses, and supporting public records help distinguish new physical locations from administrative changes at existing sites. The reporting population includes 6,971 facilities operating in 2019 and

400 classified eligible entrants in 2021–2024.[1] I hold each entry-year roster and its HRR sample fixed, following recorded service through 2024. Cohort totals are compared across treated and control HRRs within randomization regions, adjusting for 2019 market outcomes and predetermined characteristics, with parallel analyses of incumbents and the covered market. Entry can respond to the reform, so cohort totals reflect the number and composition of entrants as well as their subsequent service. These comparisons concern recorded supply in retained markets: sample inclusion depends on follow-up availability, and totals sum verified facility contributions.

Positive supply responses are concentrated among early entrants. For the 2021 cohort, the average estimates over 2021–2024 are 0.125 additional home dialysis–providing facilities and 183.01 additional Medicare home-treatment equivalents per HRR; both are statistically significant. These increases are substantial relative to control means of 0.182 facilities and 158.44 treatment equivalents. The 2022 cohort also shows a significant increase in providing facilities, although its treatment estimate is imprecise. The 2023 cohort has a significant decline in treatment volume, while both 2024 estimates are imprecise.[2] Estimates for incumbents and the covered market leave aggregate supply growth uncertain. The evidence therefore locates a positive contribution among early entrants, with limited evidence of a corresponding expansion in total regional supply.

An accounting decomposition helps interpret the early entrant response by separating contributions from original entrant counts and average service per entrant. For the 2021 cohort, entrant counts account for most of the difference in providing facilities and approximately two-thirds of the treatment difference. Average service is measured across the original roster, retaining known nonproviders and later exits in the denominator. Higher reported training and administrative staffing accompany this cohort's service expansion, while most other input estimates are imprecise. These patterns are consistent with resources devoted to establishing and supporting care at new locations. Substantial sampling uncertainty surrounds the accounting components, and the separate causal roles of entrant composition and operating decisions remain unresolved. Robustness checks examine Poisson

[1]The population covers independent and chain-owned freestanding facilities eligible to file the Independent Renal Dialysis Facility Cost Report (Form 265). Entry denotes a new physical site, including a location opened by an existing chain. The main cohort-specific HRR samples retain 362 of the 400 entrants; the 2020 transition cohort is examined separately. Section 3 describes classification and coverage.

[2]Cohort averages cover entry through 2024. Differences across cohorts combine entry-year conditions, facility age, and subsequent policy exposure.

pseudo-maximum likelihood (PPML) estimates, exclusions of the largest markets, complete facility histories, and supplementary follow-up through 2025.[3]

The paper contributes to the literature on provider payment by tracing the care supplied after facility entry. Building on evidence that payment reform affects provider entry and exit (Huckfeldt et al. 2013), I measure how successive cohorts contribute to the targeted service over the years following their establishment. Linking physical entry histories to operating records makes both active provision and treatment volume observable along these cohort paths. The distinction between entrant numbers and average service also connects the analysis to research on the subsequent performance of firm cohorts (Sedláček and Sterk 2017). Comparisons with incumbents and the covered market place these contributions in their economic context: a substantial response relative to a small cohort's initial scale can accompany an uncertain change in aggregate supply. This perspective helps distinguish where providers respond to payment incentives from the extent of overall market expansion, giving facility entry a measurable role in the evaluation of healthcare payment reform.

Sections 2–4 describe the institutional setting, data, and empirical strategy. Section 5 presents cohort supply, its decomposition, and comparisons with incumbents and the covered market. Section 6 examines training and operating inputs, and Section 7 reports robustness checks before the conclusion.

2 Institutional Setting

Home dialysis requires facilities to train patients or caregivers, arrange supplies, and provide continuing clinical support, even though treatment takes place outside the facility (Office of the Federal Register 2026). Medicare reimburses dialysis facilities through a prospective payment system that bundles routine services (Centers for Medicare & Medicaid Services 2023). The Centers for Medicare & Medicaid Services (CMS) introduced ETC to encourage home dialysis and transplantation by adjusting payments to eligible dialysis facilities and managing clinicians (Centers for Medicare & Medicaid Services 2020a).[4] The model covers incumbent facilities and eligible new sites, including locations opened by existing chains. Entry in this paper therefore refers to the opening of a physical location.

Participation was mandatory for eligible providers in selected HRRs. Within the randomized study frame, 91 of 302 HRRs were assigned to ETC and 211 to control, with assignment stratified across four Census regions (Centers for Medicare

[3]The 2021 point estimates remain positive across these comparisons, although precision is sensitive to the sample: excluding the three largest baseline markets makes both main estimates statistically insignificant. Section 7 reports the checks.

[4]The measured facility response reflects exposure to both facility and clinician incentives.

& Medicaid Services 2020a; Ji et al. 2022).[5] CMS proposed the model in July 2019, announced the final rule and geographic assignments in September 2020, and began payments in January 2021. The 2019 incumbent roster thus covers the proposal year and predates disclosure of geographic assignments.

ETC combines a temporary bonus on home-dialysis payments with performance-based adjustments to both home and in-center payments. The Home Dialysis Payment Adjustment (HDPA) increased eligible home-dialysis payments regardless of the provider's performance score. The bonus was 3 percent in 2021, 2 percent in 2022, and 1 percent in 2023, and ended on December 31, 2023 (Centers for Medicare & Medicaid Services 2020c, secs. 512.340–512.350). This temporary increase in revenue could help cover the training and continuing support required to expand home care.

The Performance Payment Adjustment (PPA) ties eligible home and in-center dialysis payments to performance, extending the incentive to a broader payment base. Its first payment period began in July 2022 and used performance measured during 2021; subsequent six-month payment periods likewise use earlier measurement windows. Home dialysis receives two-thirds of the combined score and transplant-related performance one-third. For each component, CMS uses the more favorable score from achievement relative to comparison benchmarks or improvement relative to the participant's own history (Centers for Medicare & Medicaid Services 2022).[6] Maximum facility bonuses and penalties rose from 4 and 5 percent, respectively, in the first two payment periods to 7 and 9 percent in July–December 2025. Actual adjustments depend on performance within these limits (Centers for Medicare & Medicaid Services 2025b). Figure 1 summarizes the payment schedule.

CMS pools performance across facilities owned wholly or in part by the same legal entity within an HRR, linking a site's financial incentive to care provided elsewhere in its local ownership group (Centers for Medicare & Medicaid Services 2025b).[7] Operators can expand training at existing home providers, introduce home services at previously in-center-only locations, or open new sites. New facilities

[5]HRRs with at least 20 percent of postal codes in Maryland were assigned to ETC under a separate rule and are excluded from this frame (Centers for Medicare & Medicaid Services 2020b). Figure A1 in Appendix A maps the assignment; the opening paragraph of Appendix B.2 lists the regional treatment and control counts.

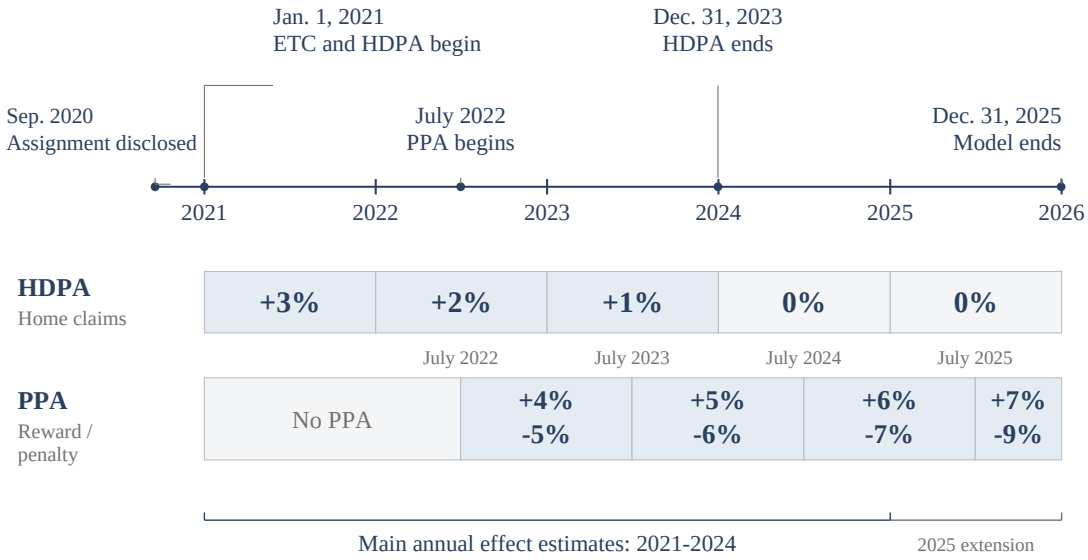
[6]The home measure uses attributed beneficiary-months, with partial credit for in-center self-dialysis and, from measurement year 3, nocturnal in-center dialysis. The transplant component combines age-adjusted waitlisting and living-donor transplantation; later periods add health-equity features (Centers for Medicare & Medicaid Services 2025b). Relating facility treatment totals to these scores requires patient attribution, denominators, and performance benchmarks.

[7]The PPA exempts ownership groups and unaffiliated facilities with fewer than 11 attributed beneficiary-years (132 beneficiary-months) per measurement period. Eligibility also depends on presence in the benchmark and measurement years and on organizational stability. CMS ownership groups may differ from the economic-parent links in the research data.

can incorporate home care into their staffing and patient-support arrangements at opening, while incumbents adjust established operations. Pooling gives operators an incentive to coordinate these choices across local facilities. Comparing entry cohorts with the fixed incumbent roster traces where supply responds; covered-market outcomes assess whether these contributions coincide with an increase in total recorded regional supply.

The payment schedule shapes the interpretation of the cohort estimates. Facilities entering in different years faced different initial bonuses and subsequent performance incentives, while also differing in facility age and market conditions. The PPA continued during 2024–2025 after the HDPA ended, so service supplied after 2023 reflects activity under ongoing performance incentives. In March 2025, CMS announced its intention to end ETC on December 31, 2025 (Centers for Medicare & Medicaid Services 2025a), potentially changing expectations about future payments during the supplementary follow-up. The estimates therefore capture responses to the combined ETC payment environment, which can affect both entry decisions and subsequent operations.[8]

Figure 1: ETC policy timing and payment phases



Notes: HDPA rates apply to eligible home-dialysis payments; PPA rates are maximum facility bonuses and penalties. Adjacent six-month periods with identical PPA limits are combined. Sources: CMS model rules and payment schedules.

[8]Separating the effects of HDPA, PPA, and investment at entry requires independent variation in these channels.

3 Data and Sample Construction

I combine public facility histories and cost reports to measure home dialysis supply in the 302 randomized HRRs from 2019 through 2024, with supplementary follow-up in 2025. HCRIS Form CMS-265-11 supplies the treatment volumes and training activity used to measure realized provision.[9] Entry counts, incumbent outcomes, market totals, and facility-level comparisons all cover facilities eligible to file this form. Reporting eligibility determines population membership; annual report availability determines which contributions are observed. Other public sources provide certification histories, patient characteristics, and geographic measures. Table A1 in Appendix A summarizes the data sources, and Figure A1 maps the randomization geography.

3.1 Population and Facility Histories

Facilities eligible to file Form 265 account for approximately 95 percent of registry-active facilities in the randomized markets. This population includes both independently owned and chain-owned facilities and excludes hospital-based units outside the reporting category; throughout the paper, "market-wide" outcomes refer to this common scope within each HRR.[10] Within this reporting population, I define the baseline roster as the 6,971 eligible facilities operating in 2019, including 264 facilities that opened that year.[11] I hold baseline membership fixed throughout follow-up, assigning later openings to separate cohorts and retaining facilities in the roster regardless of subsequent exit. The 2019 reference period precedes the disclosure of treatment assignments, although the policy proposal was already public during that year.

I classify physical entry by reconciling CMS certification and POS histories with Dialysis Facility Compare (DFC), historical addresses, and available state and operator records. The classification combines individually corroborated openings with administrative histories screened for prior operations, since certification can also change at an existing site. The reviewed file contains 876 candidate physical

[9]Form CMS-265-11 is the annual Medicare cost report for freestanding dialysis facilities, recording treatments by modality, training, staffing, revenues, and costs (Centers for Medicare & Medicaid Services 2025d).

[10]Eligibility follows certification categories and CMS reporting instructions (Centers for Medicare & Medicaid Services 2025d). Coverage measures the eligible share of registry-active facilities; treatment coverage and report completeness are assessed separately. The January 2026 register proxies year-end 2025 operating status.

[11]The roster includes facilities operating during 2019; registry coverage uses year-end counts.

entrants during 2019–2024, of which 853 fall within the Form 265 reporting scope.[12] The 264 entrants in 2019 belong to the incumbent roster. The subsequent cohorts contain 189, 149, 108, 80, and 63 eligible locations in 2020–2024, respectively. These counts describe the classified entry roster; independent corroboration of opening histories remains incomplete.

3.2 Outcomes and Observation Windows

The main outcomes measure the recorded contribution of a fixed facility population to home dialysis supply. Home dialysis–providing facilities, P_{hct}, count original cohort members with positive measured Medicare home activity in year t, and Medicare home treatments, Q_{hct}, sum their home treatment equivalents.[13] Original eligible cohort size, N_{hc}, counts locations classified as physical entrants in year c and remains fixed throughout follow-up. Known nonproviders and documented exits remain in this roster with zero activity; unresolved service remains unknown. The main incumbent and covered-market panels span 2019–2024, with average effects over 2021–2024. Each fixed entry cohort is followed from entry through 2024. The 2020 cohort provides a supplementary transition-period comparison because its formation overlaps assignment disclosure and pandemic disruption before payment implementation; its core service average includes 2020. Covered-market totals also retain verified activity from existing-site administrative successors and facilities whose physical entry year remains unresolved.

I aggregate verified facility contributions within fixed HRR samples for each population's main observation window. Where some facility outcomes remain unresolved, the aggregate sums the known activity. An empty local entry cohort contributes zero; a nonempty local population with no known contribution remains unavailable. The samples contain 301 HRRs for incumbents and the covered market, and 290, 290, 296, 293, and 295 HRRs for the 2020–2024 cohorts, respectively. Selection requires joint availability of the two main outcomes in every required year, and these HRR lists remain fixed after the entry-classification review. Geography is therefore constant within each population, although different cohorts can cover different markets. Across the 2021–2024 cohorts, 362 of the 400 classified eligible

[12]The review excludes three New York facilities acquired at existing addresses and one Florida record with evidence of prior operation at the same site from the new-location cohorts. Their verified service remains in covered-market totals. Ownership-history uncertainty is recorded separately from evidence of prior operation. The paragraph on the New York and Florida records in Appendix A gives the supporting evidence.

[13]Home treatment equivalents equal Worksheet D, column 4, lines 7 and 8, plus three times the weekly counts in lines 9 and 10. Patients can contribute multiple equivalents. Positive reported activity can establish provision when annual quantity is incomplete; home-service authorization records capability.

entrants lie in their corresponding main HRR samples.[14] The primary outcomes use observed records, documented identities and closures, and deterministic report reconciliations, excluding statistical donor-based completions. The estimates thus concern recorded supply in retained markets. Selection using follow-up availability and unreported activity remain identification concerns, discussed in Section 4.

Training treatments record work preparing patients or caregivers for home care. Training outcomes use qualifying reports from the year after entry through 2024. A selected report must last 330–400 days, cover at least 90 percent of its calendar year, and begin no earlier than entry.[15] The 2024 cohort consequently has no full-report outcome in the main window. Continuous provision requires positive home activity in every year from the first full year after entry through the observation year. The common-calendar comparison requires provision in both 2023 and 2024 and includes cohorts entering by 2023. A known zero resolves either history to zero; otherwise an unknown required year leaves the history missing. Initial configuration uses dated entry-period records, and operator composition uses documented ownership and pre-policy experience. Each additional HRR outcome retains a fixed subset of its main cohort sample with an observed aggregate in every required year; conditional facility comparisons use the observed facilities within that geography.

Operating input comparisons follow incumbents and the covered market over 2019–2024 and each fixed entry cohort from entry through 2024, with cohort averages including the entry year. They use accepted annual input fields; an entry-year report can cover a partial operating year. Inputs comprise expenses, clinical and administrative full-time equivalents (FTEs), stations, direct-care compensation, and matched ratios per all-payer treatment or patient.[16] Supplementary financial and patient-related panels end in 2024. Financial outcomes comprise Medicare payments and patient-service net income. Monetary amounts are in 2024 dollars. Clinical outcomes comprise standardized mortality, hospitalization and readmission ratios, dialysis adequacy, long-term catheter and fistula-use shares, and a vascular-access index.[17] Geographic outcomes use fixed demand weights to measure distance to the nearest documented home-capable facility, nearby facility counts, and the population share without a documented provider within 30 miles. The market all-

[14]The paragraph on sample coverage in Appendix A details the retained geography and its facility records. Annual totals can use different observed facilities within that geography.

[15]Training counts use Worksheet D, column 1, lines 3–6 and can include repeated treatments for one person. Reports may cross calendar years; totals follow the qualifying reference year. Revisions are reconciled before selection, and nonoverlapping reports can recover calendar-year totals. Identified acute-kidney-injury activity is excluded.

[16]Clinical staffing intensity is FTEs per 100 all-payer patients. Incumbent and market input totals retain 301 HRRs; matched ratios retain 300. Ratios require jointly observed components and positive denominators throughout follow-up, excluding empty local cohorts.

[17]Comparable catheter and vascular-access-index observations are unavailable in 2020.

payer home-treatment share uses matched observed reports.[18] These supplementary outcomes retain their own reporting requirements and use no statistical filling of unresolved values. Source reconciliation restores missing observations; reported FTEs, capacity records, and unresolved identity cases remain subject to source errors documented in the audit ledger.

I append available 2025 observations to the main panels for the extension in Section 7, retaining all earlier observations and the previously fixed HRR lists. Incumbent and entry-cohort service can be followed into that year. The market extension follows the verified 2024 facility roster into 2025.[19]

3.3 Baseline Adjustment and Balance

Facilities and home dialysis activity are unevenly distributed across HRRs, making market size and pre-policy supply useful adjustment variables. Table 1 reports the principal baseline characteristics and outcome levels. Additional characteristics and the eight controls appear in Table A2 of Appendix A, and Table A3 defines the study populations and variables.[20]

Every regression includes the same eight predetermined market controls, $\mathbf{X}_h$: the frozen 2019 facility-count measure of market size; patient age and female, Black, and Hispanic shares; poverty and college-education shares; and coronavirus disease 2019 (COVID-19) mortality measured around assignment disclosure. The market-size control comes from the earlier administrative baseline and totals 6,921 facilities across the 302 HRRs; it is retained as a predetermined covariate, distinct from the subsequently reconciled 6,971-facility incumbent roster.[21] Patient, socioeconomic, and pandemic controls draw on the adjustment domains in Koukounas et al. (2024); I construct HRR-level measures from public data and add market size. Patient characteristics refer to 2019, socioeconomic measures use the 2015–2019 American

[18]These supplementary market outcomes retain 299–301 HRRs. Geographic measures use documented positive capability; unknown capability remains unresolved.

[19]In 2025, both service outcomes are unavailable in 2, 4, 4, and 5 main HRRs for the 2021–2024 cohorts, respectively. Incumbent and market totals retain 301 HRRs, summing verified facility contributions. Missing values remain unfilled. The 2025 entry roster is unavailable, so 2025 entrants are excluded.

[20]Tables A4 and A5 in Appendix A report outcome distributions and reporting coverage, respectively.

[21]The count regression in the decomposition additionally adjusts for the reconciled 2019 eligible roster. Table A2 labels the frozen market-size control separately; the accompanying paragraph in Appendix A records its source.

Table 1: Pre-Policy Market Characteristics and Home Dialysis Supply

Characteristic	ETC Mean [SD]	Control Mean [SD]	Difference (SE)	Obs.
Eligible active facilities	23.37	22.80	0.52	302
	[22.52]	[30.84]	(3.16)	
Big Two facility share	75.45	76.25	−0.82	301
	[24.84]	[23.51]	(3.05)	
Other-chain facility share	16.19	14.89	1.32	301
	[20.47]	[19.11]	(2.38)	
Independent facility share	7.78	8.29	−0.50	301
	[17.00]	[17.70]	(2.16)	
Dialysis stations	349.72	272.97	72.27*	266
	[334.00]	[283.74]	(40.66)	
Medicare treatments	120.71	90.84	28.56*	266
	[117.97]	[92.59]	(14.17)	
Home-capable facilities	10.16	10.86	−0.71	302
	[9.62]	[14.93]	(1.43)	
Home dialysis–providing facilities	9.58	10.11	−0.55	302
	[9.08]	[14.03]	(1.35)	
Home-service share	42.33	45.71	−3.38*	301
	[14.45]	[18.92]	(1.99)	
Home patients	178.54	148.21	29.62	266
	[162.11]	[147.24]	(20.77)	
Home treatments	17.28	16.25	1.00	302
	[16.19]	[20.55]	(2.19)	
Treatments per providing facility	2,103.43	1,854.12	253.70	296
	[1,395.00]	[1,134.90]	(161.52)	

Notes: HRR-level statistics refer to 2019; differences adjust for randomization regions. Shares are percentages; treatment totals are thousands of equivalents; other totals are counts. Eligible active facilities are year-end counts. Home provision and treatments use verified 2019-roster contributions, and the home-service share uses the original roster denominator. Brackets report standard deviations (SD). Parentheses report standard errors (SE) with the heteroskedasticity-consistent type 2 (HC2) adjustment. Stars denote the 10, 5, and 1 percent levels (*, **, ***), based on 10,000 region-stratified randomization-inference draws.

Community Survey (ACS), and cumulative COVID-19 mortality ends on September 17, 2020.[22]

[22]Patient counts weight demographic measures. Controls are centered and scaled within randomization regions; COVID-19 mortality uses the inverse hyperbolic sine transformation. The COVID-19 cutoff is the date on the archived assignment workbook and precedes the September 18 public fact sheet; the archive does not establish the workbook's exact public release time. Outcome baselines use verified 2019 service or input measures.

4 Empirical Strategy

I estimate how ETC assignment affects supply from incumbents, the covered market, and successive entry cohorts. Incumbent estimates follow the fixed roster of eligible facilities operating in 2019, including those that entered that year. Cohort estimates measure the contribution of facilities entering in a particular year, allowing the number and composition of entrants to respond to the reform. Covered-market estimates combine activity across eligible facilities.

4.1 Market and Cohort Specifications

For HRR h in calendar year t from 2019 through 2024, I estimate outcomes for incumbent facilities using the following specification:

$$Y_{ht} = \alpha_h + \lambda_{s(h),t} + \sum_{r=2020}^{2024} \mathbf{1}\{t = r\} \left[\tau_r T_h + \boldsymbol{\gamma}_r' \mathbf{X}_h\right] + \varepsilon_{ht}. \tag{1}$$

The main outcomes Y_{ht} are the number of incumbent facilities providing home dialysis and their total Medicare home-treatment equivalents, measured at the HRR–year level. T_h indicates ETC assignment, and $s(h)$ identifies the randomization region. The HRR fixed effect α_h absorbs time-invariant market characteristics, while region-by-year effects $\lambda_{s(h),t}$ absorb annual shocks within each region, including national year effects. The treatment coefficients τ_r compare changes in treated and control HRRs relative to the omitted reference year of 2019, while the coefficients on $\mathbf{X}_h$ allow baseline characteristics to predict different annual paths. For market-wide outcomes, I apply the same specification to totals over all eligible facilities in each HRR, including later entrants. Supplementary incumbent and market operating inputs, as well as the other market outcomes, use this panel design.

I estimate the effect of ETC assignment on the total home dialysis service supplied by each fixed entry cohort using the following additive specification:

$$Y_{hct} = \lambda_{s(h),ct} + \tau_{ct} T_h + \rho_{ct} Y_{h,2019} + \boldsymbol{\gamma}_{ct}' \mathbf{X}_h + \varepsilon_{hct}. \tag{2}$$

For the core service outcomes, $c \in \{2020, \ldots, 2024\}$ denotes entry year and $t = c, \ldots, 2024$ indexes calendar years of follow-up; Y_{hct} is the recorded service total supplied by that fixed cohort in HRR h. The stratum–cohort–year intercept $\lambda_{s(h),ct}$ restricts the comparison to treated and control HRRs within the same randomization region.[23] $Y_{h,2019}$ measures baseline market supply: home dialysis–providing facilities for provision and Medicare home treatments for treatment

[23] HRR fixed effects would absorb assignment within a cohort–year comparison.

quantities and training.[24] Baseline slopes and the eight control slopes vary by cohort and observation year. If observation and sample membership are invariant to assignment, this geographic comparison identifies an intention-to-treat response encompassing changes in entry, entrant composition, and subsequent activity. Facility-average and positive-denominator ratio estimates additionally condition on the realized entrant population; equations (A1) and (A2) in Appendix B.1 specify the facility comparisons.

4.2 Functional Form, Inference, and Reporting

I estimate equations (1) and (2) by ordinary least squares (OLS), using outcomes and their baseline measures in levels. The coefficients measure changes in facilities, treatment quantities, or the other stated outcome units. This additive specification connects the supply estimates to the accounting decomposition in Section 5.2. Probability and share effects are generally expressed in percentage points; the clinical adequacy index retains its source fraction scale. As a functional-form check, Section 7 re-estimates the main supply outcomes using PPML, which accommodates zero outcomes and a multiplicative conditional mean without requiring equality of the conditional mean and variance (Santos Silva and Tenreyro 2006, 2011).[25] Table A4 in Appendix A documents the outcome distributions.

I use randomization inference (RI) to assess the sharp null of no assignment effect under the known assignment mechanism (Athey and Imbens 2017). Sparse cohort outcomes and uneven treatment volumes make this a useful complement to conventional inference, since concentrated leverage can impair large-sample approximations in randomized experiments (Young 2019). I compare a studentized efficient-score statistic with its values under 10,000 assignments preserving treatment totals within the four original regions, fitting the null nuisance model once. Stars and significance statements refer to these individual sharp-null tests; for coefficient averages, the statistic targets the corresponding weighted average of annual scores.[26] Reported standard errors use the type-two heteroskedasticity-consistent (HC2) adjustment and are clustered at the HRR level, including for facility regressions. HC2 intervals summarize uncertainty in the coefficient estimates and can differ from the RI significance assessment.

[24] Baselines cover facilities operating in 2019. Input regressions use the corresponding 2019 market total or matched ratio.

[25] PPML uses $\log(1 + Y_{h,2019})$ for cohort baselines. HRR effects absorb time-invariant baselines in the incumbent and market panels.

[26] The two-sided statistic targets a signed average and is calibrated under no effect in every included year. A zero average allowing offsetting heterogeneous effects is a different null hypothesis; see Wu and Ding (2021). Equation (A3) in Appendix B.2 defines the statistic.

Three features qualify the causal interpretation. Chains spanning treated and control HRRs may adjust service in both groups. Selection into the fixed HRR samples depends on follow-up availability, which can itself respond to ETC. Within retained HRRs, totals sum known facility contributions, so changes in reporting can affect measured supply even when geography is constant. A causal interpretation therefore requires assumptions about selection and reporting beyond random assignment. Table A5 in Appendix A reports coverage, and Section 7 compares alternative samples; high coverage and imprecise coverage differences leave uncertainty about whether missingness is ignorable. The estimates concern recorded supply in the stated samples, with extrapolation to complete market supply requiring additional assumptions. Main tables average annual coefficients equally, retaining their covariance within HRRs. Control means average the control observations included in estimation, pooling available HRR-years in the 2025 extension; adjusted counterfactual means are labeled separately in the decomposition. PPML tables report log coefficients, converted in the text using $100[\exp(\widehat{\tau}) - 1]$.

5 Main Results and Supply Decomposition

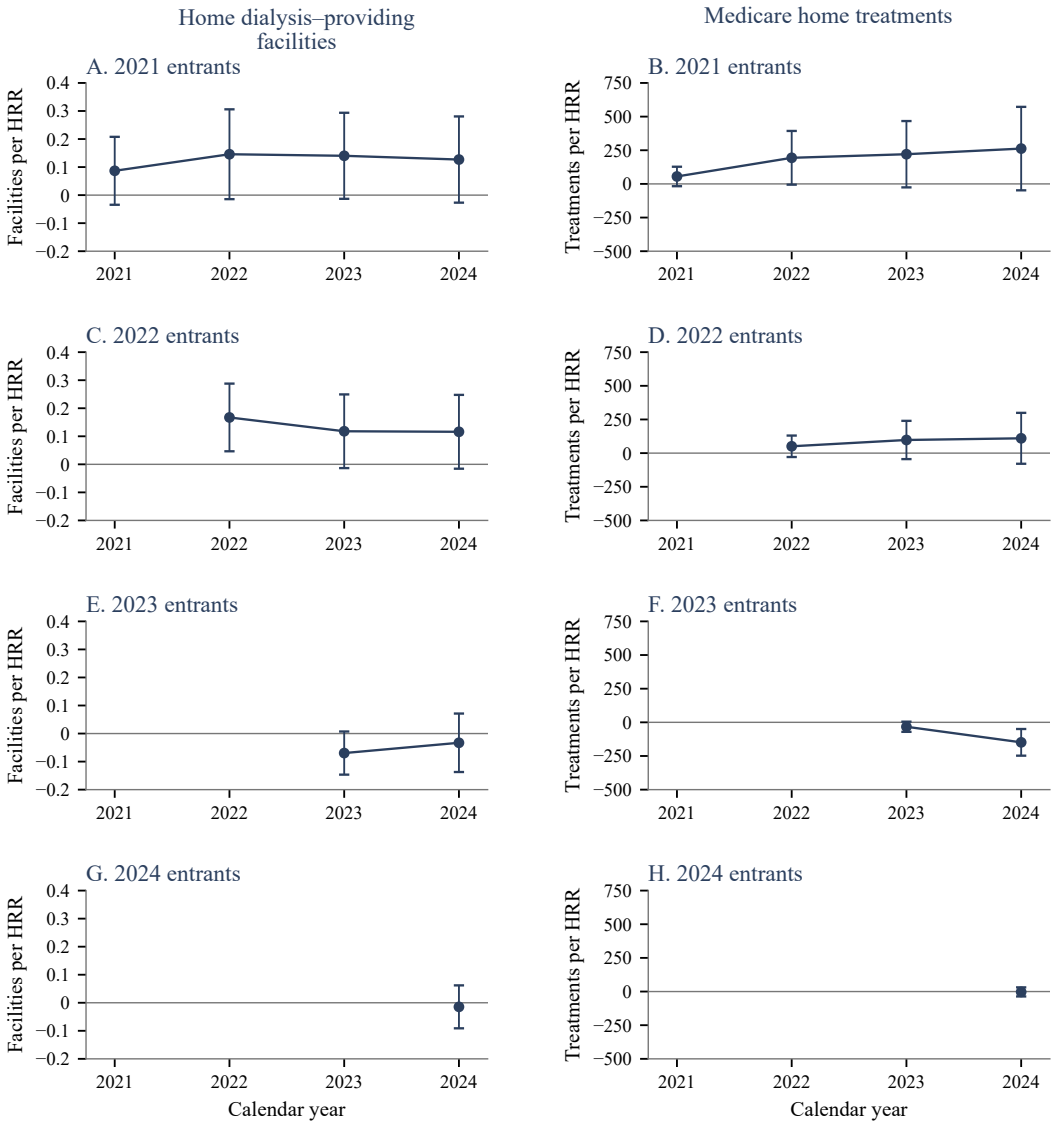
I examine the supply contributed by entry cohorts, separate it into entrant counts and average service, and compare it with incumbent and covered-market supply. Entrant outcomes combine cohort size and composition with subsequent activity. Incumbent comparisons follow a predetermined roster, while market totals assess whether the cohort responses accompany a detectable expansion of aggregate recorded supply.

5.1 Entry Cohorts and Subsequent Contributions

Positive home dialysis supply responses are concentrated among cohorts entering early in the payment period. Figure 2 traces the annual responses of the 2021–2024 entry cohorts.[27] The early cohorts generally contribute more home dialysis supply in treated markets, whereas later cohorts have weaker responses and, in some years, treatment declines. The 2021 cohort's positive treatment contribution grows over follow-up and remains visible after the temporary HDPA ends. This pattern is consistent with new facilities responding to the incentives present when they established operations, followed by continued service as the PPA remains in force. Cohort age and calendar-year conditions also vary across these paths.

[27]The 2020 cohort spans assignment disclosure and precedes payment implementation. Table A6 in Appendix C.1 collects its supplementary outcomes.

Figure 2: Home dialysis supply from fixed entrant cohorts



Notes: Annual linear estimates follow equation (2), using fixed facility rosters and cohort-specific HRRs from entry through 2024. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Table 2 reports Home dialysis–providing facilities and Medicare home treatments for each fixed 2021–2024 entry cohort.[28] For the 2021 cohort, the average estimates are 0.125 additional providing facilities and 183.01 additional Medicare home-treatment equivalents per HRR; both are statistically significant. The 2022 estimates are also positive, at 0.134 facilities and 86.12 treatments, with a significant provision response and an imprecise treatment response. For 2023, the estimates are −0.051 facilities and −90.74 treatments; the treatment decline is significant, while the provision response remains imprecise. The 2024 estimates,

[28]Averages run from entry through 2024. Table A7 in Appendix C.2 reports the annual coefficients.

−0.015 facilities and −2.71 treatments, provide little evidence of a first-year response. The positive averages are therefore concentrated among the first two payment-period cohorts, with evidence for both outcomes in 2021.

Table 2: Home Dialysis Supply from Fixed Entry Cohorts

Outcome	2021	2022	2023	2024
Home dialysis–providing facilities				
ETC effect	0.125*	0.134**	−0.051	−0.015
	(0.073)	(0.063)	(0.045)	(0.039)
Control mean	0.18	0.15	0.14	0.09
HRR-years	1,160	888	586	295
HRRs	290	296	293	295
Medicare home treatments				
ETC effect	183.01*	86.12	−90.74**	−2.71
	(102.86)	(66.15)	(33.49)	(17.23)
Control mean	158.44	87.14	93.27	32.06
HRR-years	1,160	888	586	295
HRRs	290	296	293	295

Notes: Estimates follow equation (2) on fixed cohort HRR samples. Averages weight annual coefficients equally from entry through 2024. Outcomes are providing facilities and Medicare home-treatment equivalents per HRR, summing verified contributions and known empty-cohort zeros. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. HRR-years count entry-through-2024 observations. The retained cohorts contain 135, 101, 70, and 56 facilities.

The two outcomes describe complementary dimensions of local supply. More providing facilities expand the set of entrant locations actively supporting home care, while a higher treatment total records more care delivered by the cohort. For the 2021 cohort, both responses are substantial relative to control means of 0.182 providing facilities and 158.44 treatment equivalents per HRR. Their continuation through the main follow-up is consistent with new sites establishing ongoing home-service activity.[29] The contribution to total regional supply also depends on activity elsewhere in the market. Section 7 examines model choice, influential markets, reporting restrictions, and the supplementary 2025 extension.

5.2 Entrant Counts and Average Service

I separate the cohort's supply contribution into the number of original entrants and the service supplied per entrant. For HRR h, entry cohort $c \in \{2021, \ldots, 2024\}$, and follow-up year $t = c, \ldots, 2024$, providing facilities satisfy $P_{hct} = N_{hc} p_{hct}$ and home treatments satisfy $Q_{hct} = N_{hc} \bar{q}_{hct}$ when $N_{hc} > 0$. Here N_{hc} is the original eligible cohort size, p_{hct} is the recorded proportion providing home dialysis, and $\bar{q}_{hct}$ is recorded Medicare home treatments per original entrant. Both averages

[29]Linked patient histories would distinguish additional home dialysis from transfers between providers and clarify effects on patient choice and welfare.

retain known nonproviders and later exits in the denominator.[30] They therefore describe the subsequent activity of the original cohort, including changes in its composition of active providers.

The decomposition uses the fitted values from equation (2) underlying the main cohort results. I retain the regressions for providing facilities, P, and treatments, Q, and apply the same specification to the original entrant count, N. Each count regression uses the same HRRs and equal weights as its paired service regression, including markets with zero entrants.[31] I then average predictions under each assignment over this common sample, following the regression-standardization approach described by Bartlett (2018). For outcome $K \in \{N, P, Q\}$ and assignment $d \in \{0, 1\}$, the adjusted mean is

$$\mu_{K,d,ct} = \frac{1}{|\mathcal{S}_{ct}|} \sum_{h \in \mathcal{S}_{ct}} \widehat{K}_{hct}(T_h = d). \tag{3}$$

Here $\mathcal{S}_{ct} = \mathcal{S}_c$ is the cohort's fixed main HRR sample, and $\widehat{K}_{hct}(T_h = d)$ is the fitted outcome from equation (2) with assignment set to d. Baseline outcomes and controls retain their observed values in both predictions. Averaging over the same markets places the two assignment states on a common distribution of market characteristics. These adjusted levels supply the counts and service rates needed for the decomposition: mean treatments per original entrant are $\bar{q}_{d,ct} = \mu_{Q,d,ct}/\mu_{N,d,ct}$.[32]

I apply the Shapley decomposition framework of Shorrocks (2013) to entrant counts and average service. It averages the contribution of each factor over the two possible orders of replacing control counts and service rates with their treatment counterparts. Suppressing cohort and year subscripts, the components are

$$\phi_N = (\mu_{N,1} - \mu_{N,0})\frac{\bar{q}_1 + \bar{q}_0}{2}, \qquad \phi_q = (\bar{q}_1 - \bar{q}_0)\frac{\mu_{N,1} + \mu_{N,0}}{2}. \tag{4}$$

Here ϕ_N is the treatment-volume difference assigned to original entrant counts. It multiplies the adjusted difference in cohort size, $\mu_{N,1} - \mu_{N,0}$, by the mean of the two assignment-specific service rates, $(\bar{q}_1 + \bar{q}_0)/2$. Each additional entrant is thus valued at a common treatment volume. The mean-service component, ϕ_q, multiplies the difference in treatments per original entrant by the average cohort size, $(\mu_{N,1} + \mu_{N,0})/2$. The service rate uses all original entrants as its denominator,

[30] Empty cohorts have zero totals and undefined local ratios. Numerators use verified service; denominators include every original entrant, including facilities with unresolved activity.

[31] The 2019 baselines are eligible facilities for N, providing facilities for P, and Medicare home treatments for Q. All regressions include region effects and $\mathbf{X}_h$, with slopes varying by cohort and year.

[32] Taking ratios of adjusted means retains HRRs with no entrants. In the linear model, $\mu_{K,1,ct} - \mu_{K,0,ct} = \widehat{\tau}^K_{ct}$, the assignment coefficient for outcome K.

including known nonproviders and later exits. Both components are measured in treatment equivalents per HRR: a positive value contributes to higher recorded treatment volume under ETC, and opposite signs indicate offsetting contributions. They sum to the adjusted treatment difference, $\phi_N + \phi_q = \mu_{Q,1} - \mu_{Q,0}$, assigning half the interaction between count and service changes to each factor.

For providing facilities, define the adjusted provision proportion as $p_d = \mu_{P,d}/\mu_{N,d}$. The corresponding count and provision-rate components are

$$\phi_N^P = (\mu_{N,1} - \mu_{N,0})\frac{p_1 + p_0}{2}, \qquad \phi_p^P = (p_1 - p_0)\frac{\mu_{N,1} + \mu_{N,0}}{2}. \tag{5}$$

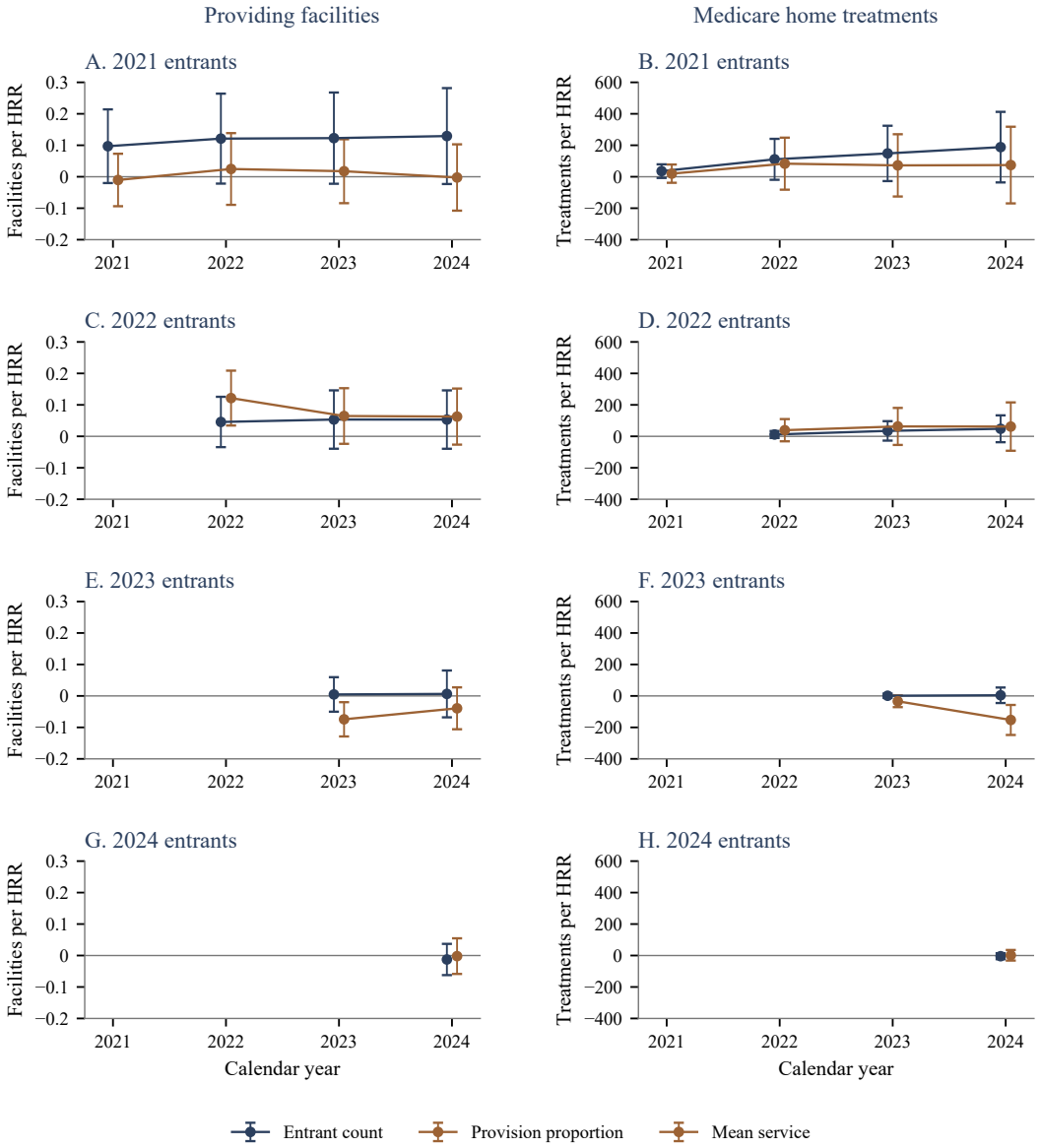
The superscript P identifies the providing-facility outcome. Its count component, ϕ_N^P, values the difference in original entrant counts at the average provision proportion, $(p_1 + p_0)/2$. The provision-rate component, ϕ_p^P, values the difference in the share of original entrants providing home dialysis at the average cohort size. Both are measured in providing facilities per HRR and sum to $\mu_{P,1} - \mu_{P,0}$. A larger cohort can therefore contribute more providing facilities even with an unchanged provision proportion; a lower proportion can offset part of that count contribution. Known nonproviders and exits remain in the original denominator.[33]

Figure 3 shows the two annual components for both supply outcomes. Their sum equals the corresponding cohort and year effect in Figure 2. The 2021 cohort's positive providing-facility response is largely assigned to original entrant counts, with smaller changes in the provision proportion. Its treatment response has positive count and mean-service components throughout follow-up, and the count component rises over time. The 2022 provision response places more weight on the provision proportion, whereas the 2023 treatment decline is concentrated in mean service. Later cohorts have shorter observed histories, and the intervals leave the relative component sizes uncertain.

Table 3 quantifies these allocations. For the 2021 cohort, Panel A assigns 0.118 of the total 0.125 additional providing facilities per HRR to original entrant counts and 0.007 to the provision proportion, approximately 94 percent and 6 percent of the point allocation. Panel B assigns 120.7 of the 183.0 additional treatment equivalents to counts and 62.3 to mean service, approximately 66 percent and 34 percent. Both treatment components have substantial sampling uncertainty. The allocations describe different aspects of the same cohort: additional provider locations are mainly associated with cohort size, while average service accounts for a larger share of the treatment difference. For 2022, the provision-proportion component is 0.083 of a total 0.134 facilities; its two treatment components are positive. The

[33]Annual components are averaged over each cohort's follow-up. Standard errors use 2,000 HRR bootstrap draws, stratified by region and assignment and shared across outcomes and years.

Figure 3: Entrant counts and average service: annual accounting components



Notes: Components from equations (4)–(5) sum to the annual cohort effects in Figure 2. Bars are estimates plus or minus 1.96 bootstrap standard errors, using 2,000 HRR draws stratified by region and assignment and shared across outcomes and years.

2023 treatment decline is almost entirely assigned to mean service, and the 2024 differences are small. Panel C reports the adjusted counts, service totals, and rates entering the calculation. All 2,000 bootstrap draws for the 2021 treatment allocation satisfy the economic bounds, while 781 draws for 2023 contain a negative fitted quantity, limiting a literal service-rate interpretation of that allocation. Table A8 in Appendix C.3 reports the uncertainty and bounds diagnostics; its opening paragraph specifies the resampling procedure and admissibility criteria.

The components describe accounting contributions with several possible economic sources. The provision proportion can change through entrant composition, subsequent adoption, or cessation of home service; mean treatments can also

Table 3: Entrant Counts and Average Service: Two-Factor Allocations

Component	2021	2022	2023	2024
Panel A. Home dialysis–providing facilities				
Entrant count	0.118	0.051	0.006	−0.013
	(0.071)	(0.045)	(0.033)	(0.025)
Provision proportion	0.007	0.083	−0.057	−0.002
	(0.050)	(0.042)	(0.029)	(0.029)
Total difference	0.125	0.134	−0.051	−0.015
	(0.071)	(0.062)	(0.043)	(0.037)
HRRs per year	290	296	293	295
HRR-years	1,160	888	586	295
Panel B. Medicare home treatments				
Entrant count	120.7	31.4	2.7	−4.3
	(72.9)	(28.6)	(16.1)	(9.7)
Mean service per original entrant	62.3	54.7	−93.4	1.6
	(81.8)	(54.7)	(32.3)	(17.2)
Total difference	183.0	86.1	−90.7	−2.7
	(99.9)	(65.1)	(33.2)	(17.0)
HRRs per year	290	296	293	295
HRR-years	1,160	888	586	295
Panel C. Adjusted levels (control / ETC)				
Original entrants per HRR	0.391 / 0.644	0.314 / 0.403	0.235 / 0.247	0.198 / 0.171
Providing facilities per HRR	0.179 / 0.304	0.143 / 0.277	0.138 / 0.087	0.096 / 0.081
Treatments per HRR	163.0 / 346.0	87.0 / 173.1	98.3 / 7.6	31.8 / 29.1
Provision proportion	0.457 / 0.471	0.455 / 0.686	0.588 / 0.352	0.485 / 0.474
Treatments per original entrant	416.6 / 536.9	277.0 / 429.5	417.7 / 30.6	160.8 / 169.7

Notes: Annual Shapley allocations use equations (3)–(5) and are averaged equally from entry through 2024. Components sum to the corresponding effects in Table 2. Service measures retain verified contributions; entrant counts retain the original roster. Parentheses report standard errors from 2,000 HRR bootstrap draws, stratified by region and assignment and shared across outcomes and years. These components describe accounting allocations. Panel C gives adjusted control and ETC levels from equation (3), averaged over follow-up. These differ from the observed control means in Table 2.

reflect patient mix and treatment intensity.[34] Causal mediation through entry and operating decisions remains unresolved. This distinction parallels the separation of firm numbers and average firm size in Sedláček and Sterk (2017); their analysis of unobserved growth potential additionally relies on a structural model.

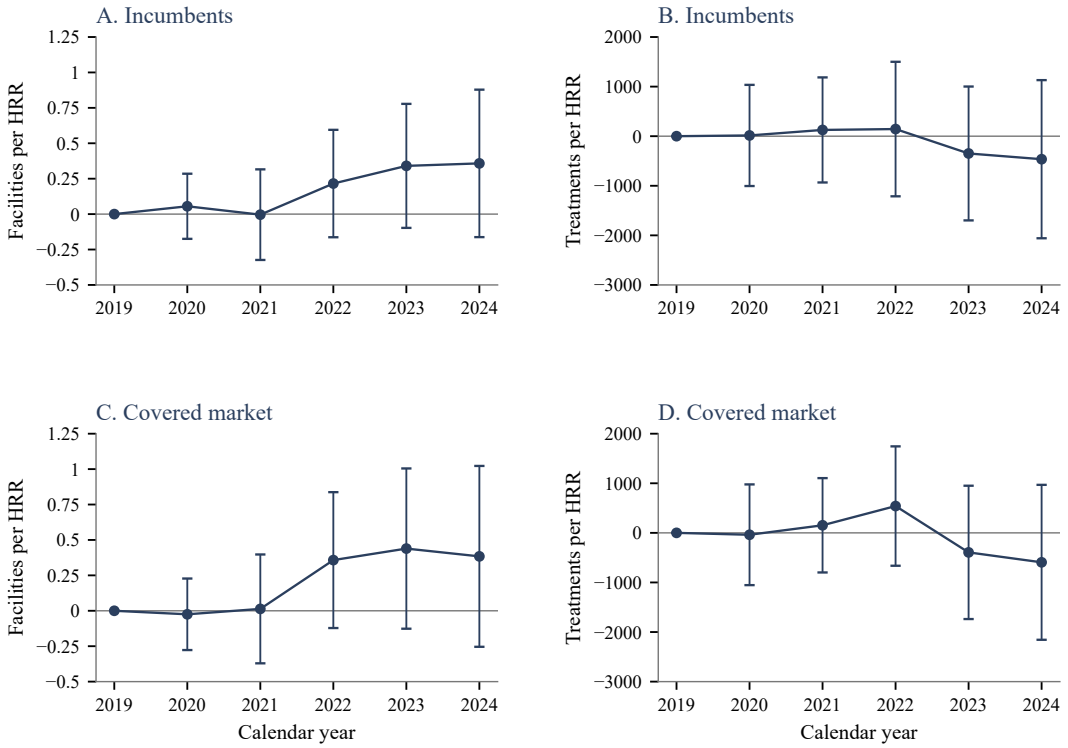
5.3 Incumbent and Market-Wide Home Dialysis Supply

Figure 4 places the entrant response alongside incumbent and covered-market supply.[35] Both populations show modest positive provider-count estimates in the later years, while treatment-volume estimates turn negative toward the end of the main window. The wide intervals provide limited evidence of sustained expansion across the broader market, despite the positive contribution from early entrants.

[34]Table A9 in Appendix C.4 reports continued provision; Table A10 in Appendix C.5 reports initial configuration and operator composition.

[35]Both panels retain 301 HRRs; the facility records contributing to annual totals can change. Table A7 in Appendix C.2 reports annual estimates.

Figure 4: Incumbent and covered-market home dialysis supply



Notes: Annual linear estimates follow equation (1), with 2019 as the reference year. Both populations retain 301 HRRs during 2019–2024 and sum verified facility contributions. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Table 4 summarizes the average responses. Panel A follows the original incumbent roster and reports 0.228 additional providing facilities and 135.62 fewer Medicare home-treatment equivalents per HRR. Panel B includes the broader eligible reporting population, with 0.299 additional providing facilities and 73.52 fewer treatments. All four averages are statistically insignificant, leaving both modest increases and decreases in aggregate supply plausible. The cohort estimates locate a positive recorded contribution among early entrants, while the net response across established facilities and the covered market remains unresolved.[36]

The scale of the entry cohorts helps explain this contrast. In the common accounting samples, the 2021–2024 cohorts account for 4.0 percent of providing facilities and 2.2 percent of Medicare home treatments in control markets.[37] Substantial changes within these small cohorts can therefore make a modest contribution to total regional supply. Table A11 in Appendix C.6 compares market components under a common linear design. Its residual category includes administrative succes-

[36]Cohort and incumbent models use different baseline adjustments. Table A11 in Appendix C.6 reports additive contributions under a common linear specification and sample.

[37]Shares pool matched control HRR-years in 2021–2024: 271 HRRs for provision and 260 for treatments. The 2021 cohort accounts for 2.0 and 1.2 percent, respectively.

Table 4: Incumbent and Covered-Market Home Dialysis Supply

Outcome	Home dialysis–providing facilities	Medicare home treatments
Panel A. Fixed 2019 incumbent roster		
ETC effect	0.228	−135.623
	(0.192)	(640.823)
Control mean	10.01	13,336.65
HRR-years	1,204	1,204
HRRs	301	301
Panel B. Covered market		
ETC effect	0.299	−73.524
	(0.243)	(579.568)
Control mean	10.82	14,122.39
HRR-years	1,204	1,204
HRRs	301	301

Notes: Estimates follow equation (1), relative to 2019, on 301 fixed HRRs. Averages weight the 2021–2024 coefficients equally. Outcomes are providing facilities and Medicare home-treatment equivalents per HRR, summing verified facility contributions. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. Each equation uses 1,806 HRR-years over 2019–2024; the average-window count is 1,204.

sors and facilities without an assigned physical opening year and accounts for much of the negative treatment point estimate in that sample. Patient reallocation and the role of unresolved entry histories consequently remain open explanations. The evidence supports an early-entrant contribution without establishing a net market expansion. Section 6 examines the training and operating inputs accompanying that contribution.

6 Additional Outcomes and Economic Implications

Additional outcomes describe the training and operating resources accompanying the supply responses of the 2021–2024 entrant cohorts. They cover training activity, operating expenses, dialysis stations, clinical and administrative staffing, direct-care compensation, and cost and staffing intensity. Supplementary comparisons examine incumbent and market-wide inputs, financial and clinical outcomes, geographic availability, and the all-payer home-treatment share.

Table 5: Additional outcomes in fixed entry cohorts: average responses

Outcome / statistic	2021	2022	2023	2024
Panel A. Reported training activity				
Training treatments per HRR				
ETC effect	25.99*	29.51	−28.58**	—
	(14.00)	(20.22)	(12.09)	—
Control mean	20.97	16.93	28.81	—
HRR-years	864	590	292	—
HRRs	288	295	292	—
Panel B. Operating-input totals				
Operating expenses (million dollars per HRR)				
ETC effect	0.462	0.075	−0.092	0.007
	(0.329)	(0.116)	(0.133)	(0.057)
Control mean	0.816	0.496	0.344	0.161
HRR-years	1,140	879	572	290
HRRs	285	293	286	290
Dialysis stations per HRR				
ETC effect	0.691	0.130	−0.229	−0.508
	(0.912)	(0.535)	(0.472)	(0.392)
Control mean	2.233	1.461	0.742	0.791
HRR-years	1,140	873	568	289
HRRs	285	291	284	289
Clinical FTEs per HRR				
ETC effect	1.500	0.214	−0.189	−0.392
	(0.980)	(0.330)	(0.301)	(0.361)
Control mean	2.370	1.343	0.809	0.689
HRR-years	1,140	879	572	290
HRRs	285	293	286	290
Administrative and management FTEs per HRR				
ETC effect	0.378*	0.055	−0.008	−0.114*
	(0.220)	(0.104)	(0.096)	(0.059)
Control mean	0.558	0.379	0.240	0.173
HRR-years	1,140	879	572	290
HRRs	285	293	286	290
Direct-care compensation (million dollars per HRR)				
ETC effect	0.127	0.036	−0.023	0.000
	(0.089)	(0.034)	(0.031)	(0.014)
Control mean	0.232	0.124	0.092	0.040
HRR-years	1,140	876	572	290
HRRs	285	292	286	290
Panel C. Matched input ratios				
Expenses per treatment (dollars)				
ETC effect	5.26	−200.78*	−35.05	−554.08
	(106.45)	(119.32)	(302.93)	(662.36)
Control mean	569.45	762.01	1,028.77	1,737.51
HRR-years	272	189	76	42
HRRs	68	63	38	42
Clinical FTEs per 100 patients				
ETC effect	1.52	−0.43	1.88	−5.90
	(1.31)	(1.55)	(4.26)	(4.03)
Control mean	10.96	11.59	11.71	16.03
HRR-years	272	183	74	40
HRRs	68	61	37	40

Continued on next page

Table 5: Additional outcomes in fixed entry cohorts: average responses (continued)

Outcome / statistic	2021	2022	2023	2024
Treatments per clinical FTE				
ETC effect	−523.40	−173.06	−482.36*	66.92
	(406.16)	(167.80)	(291.24)	(137.70)
Control mean	1,085.16	955.39	1,120.57	481.06
HRR-years	272	189	76	40
HRRs	68	63	38	40

Notes: Equal-weight cohort averages use equation (2) on fixed outcome-specific HRRs. All-payer training uses qualifying full reports after entry through 2024; inputs cover entry through 2024, allowing partial entry-year reports. Baselines are 2019 home treatments for training and the corresponding input otherwise. Totals retain partial facility coverage; ratios require matched components and positive denominators. Monetary totals are millions of 2024 dollars; per-treatment expenses are 2024 dollars. Panel C uses all-payer denominators. FTE denotes full-time equivalent; administrative staffing includes management. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. Dashes indicate unavailable outcomes.

6.1 Reported Training Activity

Reported training rises for the 2021 cohort and falls for the 2023 cohort, paralleling their home-treatment responses (Panel A of Table 5). The average 2021 response is 25.99 additional training treatments per HRR, approximately 124 percent of the control mean of 20.97, and is statistically significant. The 2022 estimate is also positive, at 29.51 treatments, but imprecise. For 2023, the estimate is a significant decline of 28.58 treatments per HRR, compared with a control mean of 28.81.[38] These treatments record work preparing patients or caregivers to perform dialysis at home. The 2021 increase is consistent with greater preparation accompanying that cohort's larger home-service contribution, while the 2023 decline accompanies lower recorded treatment volume. Regional training totals depend on the number of entrants, the number of people trained per facility, and the training received per person, so both cohort size and activity within facilities can contribute to the response.[39] Figure A2 in Appendix D.1 displays the annual HRR training paths.[40]

[38]Training follow-up begins the year after entry (Section 3.2), leaving no main-window observation for 2024 entrants.

[39]Unique trainees, completed courses, and linked patient histories are unobserved, limiting conclusions about recruitment and successful home-dialysis adoption.

[40]Table A12 in Appendix D.1 reports HRR and facility estimates, including the 2020 cohort; Table A5 in Appendix A reports coverage. Facility comparisons condition on realized entry and report availability.

6.2 Operating Inputs

The 2021 cohort also has positive estimated responses in operating expenditure, physical capacity, and employment (Panel B of Table 5).[41] The average estimates amount to an additional $0.462 million in operating expenses, 0.691 dialysis stations, 1.500 clinical FTEs, 0.378 administrative and management FTEs, and $0.127 million in direct-care compensation per HRR. The administrative and management response is statistically significant; the other four averages are imprecisely estimated. These estimates are consistent with greater operating resources accompanying the cohort's larger service contribution, although the amounts remain uncertain. Because HRR totals combine entrant numbers and composition with resources used per facility, they measure the aggregate operating scale of the cohort.

The administrative staffing increase amounts to approximately 68 percent of the control mean of 0.558 FTEs. It is consistent with additional labor devoted to organizing and managing the 2021 cohort's facilities. More entrant locations can raise these requirements, alongside changes in staffing within each facility.[42] For the 2024 cohort, administrative and management staffing falls by 0.114 FTEs, approximately 66 percent of the control mean of 0.173; this decline is also statistically significant. All remaining input-total averages for the 2022–2024 cohorts are statistically insignificant. The administrative estimates thus have opposite signs for the earliest and latest cohorts, while changes in capacity, clinical employment, and expenditure remain less precisely determined. Annual expense and station paths appear in Figure A3, staffing in Figure A4, and compensation in Figure A5, all in Appendix D.2.

6.3 Cost and Staffing Intensity

Panel C of Table 5 compares cost and staffing ratios in nonempty local cohorts with jointly observed components and positive denominators throughout follow-up. These conditional samples are substantially narrower than the samples for totals.[43] Operating expenses per treatment fall among the observed 2022 entrants, while most other ratios are imprecisely estimated. The average reduction is $200.78 per treatment, approximately 26 percent of the control mean of $762.01, and is statistically significant; the corresponding averages for the other three cohorts are

[41] Each input uses a fixed subset of the cohort's main HRRs and its corresponding 2019 market baseline in equation (2). Totals retain verified partial contributions; Section 3.2 specifies reporting rules.

[42] The estimates leave the role of administrative requirements in entry decisions unresolved. Resources shared across dialysis modalities also limit conclusions about home-dialysis expenditure and efficiency.

[43] The 2021 totals use 285 HRRs, compared with 68 for matched ratios; later-cohort ratios use 37–63. Ratios require jointly observed components and positive denominators, excluding empty cohorts. Differences in samples and reporting windows limit cross-outcome comparisons.

statistically insignificant. This ratio relates operating expenditure to all-payer treatment volume. It can fall as treatment volume expands relative to operating outlays or as the composition of entrants and services changes. The result is consistent with lower recorded expenditure per unit of care among the observed 2022 entrants, conditional on realized entry and reporting under each assignment.[44]

The labor ratios relate clinical staffing to patient numbers and treatment volume. Average responses in clinical FTEs per 100 patients are statistically insignificant in all four cohorts; for 2021 entrants, the estimate is an additional 1.52 FTEs per 100 patients relative to a control mean of 10.96. Treatments per clinical FTE fall by 482.36 for the 2023 cohort, approximately 43 percent of the control mean of 1,120.57, with the decline statistically significant. The 2021 estimate is also negative, at 523.40 fewer treatments per clinical FTE, but imprecise, as are the 2022 and 2024 averages. The 2023 response records lower treatment volume per unit of clinical labor and may reflect changes in service volume, care requirements, or entrant composition. Interpreting this ratio as productivity, or linking staffing intensity to workload and quality, requires information on patient needs and services delivered. Figure A5 reports the annual cost-per-treatment paths, and Figure A6 reports staffing intensity, both in Appendix D.2.[45]

7 Robustness Checks

I examine sensitivity to functional form, influential markets, and facility reporting, followed by the supplementary 2025 extension. The 2021 contribution remains positive across these comparisons, while its precision depends on the sample.

7.1 Functional Form

Figures A7 and A8 in Appendix E.1 show the annual PPML responses for cohorts and for incumbent and covered-market supply. The early cohorts generally have positive estimates, the 2023 treatment path is negative, and aggregate responses remain uncertain. PPML allows assignment to change the conditional mean proportionally.[46] The 2021 comparison retains the same 290 HRRs under both functional forms. For 2023, separation reduces fitted support from 293 to 211

[44]Assessing profitability or patient financial gains also requires facility revenues and patient payments.

[45]Table A13 in Appendix D.2 reports incumbent and market input responses; Table A14 gives annual cohort estimates. Table A6 in Appendix C.1 covers 2020 entrants, and Table A15 in Appendix D.3 reports the supplementary financial, clinical, geographic, and all-payer-share comparisons.

[46]The specification starts from the main panels, retains assignment and the eight controls, and transforms cohort baselines as described in Section 4.

HRRs, so its comparison combines functional form, baseline transformation, and sample composition.

Table A16 in Appendix E.1 reports the annual and average estimates. The 2021 provider and treatment coefficients are 0.644 and 0.781, corresponding to increases of 90.4 and 118.4 percent; both are statistically significant.[47] The 2022 provider increase and 2023 treatment decline are statistically significant. Other cohort averages and all incumbent and market averages are imprecise. This agreement supports the direction of the early-cohort response under a multiplicative mean, while differences in scales and fitted support limit direct comparisons of magnitudes.

7.2 Influential Markets and Facility Reporting

Table A17 in Appendix E.2 compares the main estimates with restrictions on market size and facility reporting. Markets are ranked by their reconciled 2019 eligible-facility counts. Excluding the largest market leaves the 2021 estimates at 0.127 additional providing facilities and 181.16 additional Medicare home-treatment equivalents per HRR, both statistically significant. Excluding the three largest reduces them to 0.117 facilities and 160.12 treatments, with neither statistically significant. The 2022 provider increase and 2023 treatment decline remain significant under both exclusions. Positive 2021 point estimates therefore persist beyond the largest markets, while their precision is sensitive to those observations.

The complete-history estimates in Table A17 require both main outcomes to be resolved for every original facility in every required year.[48] The 2021 cohort retains 281 of 290 HRRs, with estimates of 0.106 providing facilities and 172.54 treatment equivalents. The treatment increase is statistically significant; the provider estimate is imprecise. The 2022 provider increase and the 2023 treatment decline are also statistically significant. Incumbent and market samples fall from 301 to 97 and 86 HRRs, respectively, with their averages remaining imprecise. These restrictions test sensitivity to complete reporting in substantially selected samples; they leave the unreported contribution in excluded markets and the process governing missingness unresolved.

7.3 Supplementary Follow-Up in 2025

Figures A9 and A10 in Appendix E.3 extend the paths in Figures 2 and 4. The 2021 cohort retains positive provider and treatment point estimates, the 2023 treatment decline persists, and the aggregate paths remain uncertain. I append available

[47] Percentage responses apply $100[\exp(\widehat{\tau}) - 1]$ after averaging annual log coefficients. The supplementary 2020 core-service average covers 2020–2024 in both functional forms.

[48] Documented physical exits contribute zeros; unresolved reports remain missing.

2025 observations and re-estimate the full annual specifications, retaining all earlier observations, main HRR lists, and baseline adjustments.[49] This extra year includes the early-termination announcement, so the extension combines further cohort aging with a possible change in expectations about future payments.

Table A18 in Appendix E.3 reports the full annual estimates and extended averages. For the 2021 cohort, the averages are 0.131 additional providing facilities and 210.63 additional Medicare home-treatment equivalents per HRR, both statistically significant, although several annual estimates remain imprecise. The 2022 provider average is statistically significant, while its treatment response is imprecise; the 2023 treatment average is −123.04, statistically significant. Incumbent averages are 0.248 facilities and −181.19 treatments, and market averages are 0.315 facilities and −167.42 treatments. All four are statistically insignificant. The extension supports continued recorded activity among early entrants, while net regional expansion and patient reallocation remain unresolved.

8 Conclusion

ETC's geographic randomization provides evidence on how payment reform affects the services supplied by new facilities after entry. The 2021 cohort supplies an estimated 0.125 additional home dialysis–providing facilities and 183.01 additional Medicare home-treatment equivalents per HRR on average through 2024, with both responses statistically significant. The 2022 provision response is also positive, while the 2023 treatment response is negative. Incumbent and covered-market estimates remain imprecise, leaving net regional expansion uncertain. PPML and supplementary 2025 follow-up preserve this pattern, although the precision of the 2021 estimates is sensitive to excluding the three largest baseline markets.

The decomposition assigns most of the 2021 provider-count difference and approximately two-thirds of its treatment difference to original entrant counts. Higher reported training and administrative staffing accompany the cohort's larger service contribution, consistent with new locations developing the capacity to support home dialysis. Uncertainty around the accounting components and limited evidence on changes per entrant leave the roles of composition and operating behavior unresolved. Separating incentives at entry from subsequent exposure and facility age requires additional variation.

Following entrants alongside incumbents and the covered market helps locate the recorded supply response to payment reform. These estimates concern freestanding facilities through 2024 and remain qualified by sample selection using follow-up availability, incomplete reports, and unresolved physical entry histories. More

[49]The market extension follows the verified 2024 roster. Section 3.2 specifies the 2025 gaps.

complete reports and linked patient histories would help distinguish additional home-dialysis use from transfers across providers. Assessing access and welfare also requires evidence on resource costs and patient outcomes.

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A Data, Measurement, and Coverage

Table A1 identifies the sources used to construct facility populations and outcomes, and Figure A1 displays the randomization geography. The provenance archive records report identifiers, worksheet cells, dates, and acceptance decisions, distinguishing outcome measurements from records used only to corroborate a facility's identity.

Table A1: Data Sources and Measurement Roles

Source	Measurement role
ETC assignment and rules	CMS HRR assignment workbook (September 17, 2020), final rule, and subsequent model documents; 302 HRRs in four Census regions. CMS ETC materials.
HRR geography	Dartmouth Atlas 2019 ZIP–hospital service area–HRR crosswalk (`ZipHsaHrr19.csv.zip`), held fixed across years. Dartmouth supplemental files.
Facility registry and identity	Provider of Services (POS) snapshots from the Quality Improvement and Evaluation System (QIES), supplemented by Internet QIES (iQIES) for the 2026 transition: identifiers, dates, addresses, categories, services, staffing, and stations. Snapshot and download dates are recorded separately. CMS POS files.
Service reports	Healthcare Cost Report Information System (HCRIS) Form CMS-265-11 all-years renal release, with headers and worksheet cells linked by report identifier and audited period mappings. CMS cost reports; all-years archive.
Reporting instructions	Provider Reimbursement Manual, Part 2, Chapter 42, using the instructions applicable to each report period. May 13, 2025 instructions, Transmittal 11.
Public facility reports	Dialysis Facility Compare (DFC) releases and public Dialysis Facility Reports (DFRs), with release dictionaries and measurement-year mappings. The fiscal-year 2022 DFR supplies the 2019 demographic controls. CMS dialysis data.
Community controls	2015–2019 American Community Survey (ACS) five-year tables B17001 (poverty) and B15003 (education), aggregating numerators and denominators through fixed geography. Census summary files.
COVID-19 around disclosure	Dartmouth HRR cumulative deaths per 100,000 through September 17, 2020, transformed by the inverse hyperbolic sine before covariate standardization. Dartmouth COVID-19 series.
Geographic access	Census Geocoder and ZIP Code Tabulation Area (ZCTA) Gazetteer coordinates; Dartmouth crosswalk; fixed population and rural classifications; chronic kidney disease (CKD) estimates from the Centers for Disease Control and Prevention (CDC) Population Level Analysis and Community Estimates (PLACES), 2022 ZCTA release, based on the 2020 Behavioral Risk Factor Surveillance System (BRFSS). This demand weight partly overlaps the 2020 transition year. CDC PLACES; Census Gazetteer.

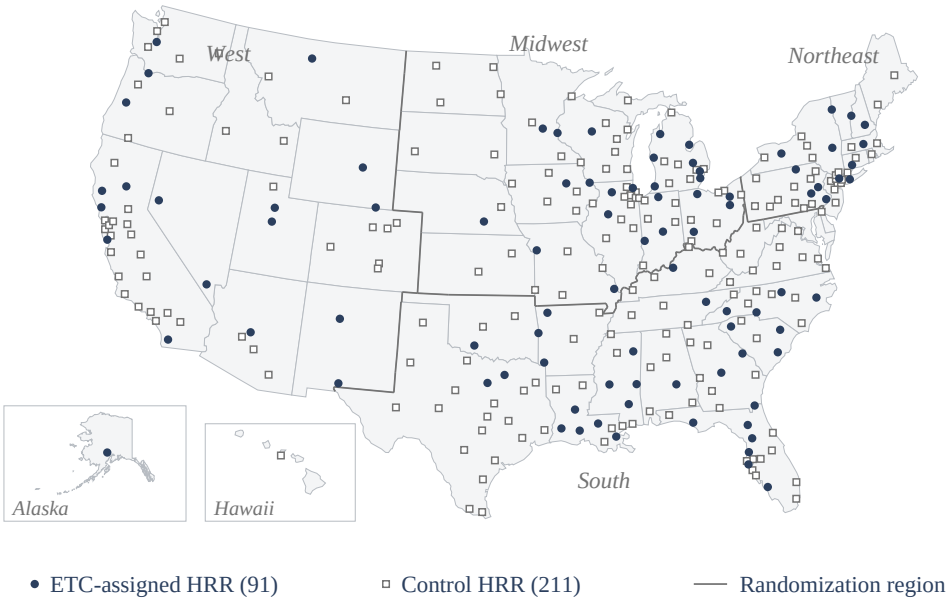
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Table A1: Data Sources and Measurement Roles (continued)

Source	Measurement role
Prices	Bureau of Labor Statistics (BLS) Consumer Price Index for All Urban Consumers (CPI-U), annual U.S. city average, all items; monetary outcomes use 2024 dollars. BLS price data.
Entry and operator verification	National Plan and Provider Enumeration System (NPPES) records and the National Provider Identifier (NPI) Registry; Medicare enrollment records; state licensing and certificate-of-need records; Securities and Exchange Commission (SEC) Electronic Data Gathering, Analysis, and Retrieval (EDGAR) filings; and dated operator announcements. These distinguish physical openings from moves, renumbering, and ownership changes; ownership at entry requires dated supporting evidence.
Dated configuration supplements	North Carolina State Medical Facilities Plans and Illinois annual end-stage renal disease (ESRD) facility profiles provide explicitly dated station counts. California's System for Integrated Electronic Reporting and Auditing (SIERA) reports and early HCRIS reports supply supplemental period measures; opening configuration requires a dated observation. Facility-specific provenance preserves the document, page or field, observation date, and acceptance decision.

Notes: Linkage preserves facility identity, report periods, worksheet definitions, and acceptance decisions. The provenance archive distinguishes measurements from corroborating evidence. The assignment workbook is dated September 17, 2020; the public CMS fact sheet is dated September 18.

Figure A1: Geographic assignment of the ETC model



Notes: Points locate the 302 randomized HRRs: 91 ETC and 211 control. Dark boundaries delineate randomization regions; light boundaries delineate states. Alaska and Hawaii are inset.

Table A2 reports additional baseline characteristics and the eight controls. Retrospective 2019 patient characteristics from the fiscal-year 2022 public dialysis report are weighted by patient counts within HRR; community measures use aggregated ACS numerators and denominators. The COVID-19 cutoff is September 17, 2020. The frozen market-size covariate comes from the earlier Form 265 administrative baseline, totaling 6,921 facilities, compared with 6,971 in the reconciled incumbent roster; the latter count also supplies the outcome baseline for entrant-count regressions. The measures have different data vintages and remain distinct regressors. The source archive retains their HRR-level reconciliation. Table A3 defines the populations and outcomes.

Table A2: Additional Baseline Characteristics and Predetermined Controls

Characteristic	ETC Mean [SD]	Control Mean [SD]	Difference (SE)	Obs.
Panel A. Eight predetermined controls				
Frozen 2019 facility count (market-size control)	23.34	22.73	0.56	302
	[22.43]	[30.72]	(3.15)	
Patient mean age	62.71	63.13	-0.41^{**}	302
	[1.81]	[1.75]	(0.19)	
Female patients	42.55	42.14	0.40	302
	[3.36]	[2.98]	(0.36)	
Black patients	29.49	25.14	4.22^{*}	302
	[23.78]	[21.49]	(2.26)	
Hispanic patients	10.40	14.97	-4.58^{***}	302
	[13.13]	[18.05]	(1.68)	
Community poverty rate	14.31	14.11	0.18	302
	[3.80]	[4.05]	(0.41)	
Community college-or-higher share	28.39	28.69	−0.28	302
	[7.58]	[8.42]	(0.91)	
COVID-19 deaths by Sept. 17, 2020	56.16	52.01	3.99	302
	[44.57]	[49.35]	(5.25)	
Panel B. Treatment supply				
All dialysis patients	2.13	1.69	0.43	266
	[2.12]	[1.75]	(0.26)	
All dialysis treatments	216.23	171.66	42.92	266
	[219.46]	[176.47]	(26.77)	
In-center patients	1.97	1.55	0.41^{*}	263
	[1.97]	[1.62]	(0.24)	
In-center treatments	188.00	148.21	38.24	266
	[194.88]	[155.29]	(23.64)	
All-payer home-treatment share	13.79	14.51	−0.67	265
	[4.90]	[6.85]	(0.73)	
Panel C. Operating inputs and financial outcomes				
Clinical FTE	236.61	187.37	47.47	266
	[245.35]	[196.26]	(29.87)	

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Table A2: Additional Baseline Characteristics and Predetermined Controls (continued)

Characteristic	ETC Mean [SD]	Control Mean [SD]	Difference (SE)	Obs.
Administrative FTE	46.53	37.01	9.07	266
	[47.24]	[37.10]	(5.69)	
Clinical FTE per 100 patients	11.090	11.088	0.022	265
	[1.882]	[2.147]	(0.255)	
Treatments per clinical FTE	925.95	932.30	−7.11	265
	[115.16]	[122.95]	(15.40)	
Total expense	69.17	55.12	13.59	266
	[70.67]	[57.29]	(8.60)	
Expense per treatment	325.44	322.13	3.99	265
	[37.82]	[36.90]	(4.53)	
Total Medicare payment	46.45	35.37	10.67*	266
	[46.33]	[35.97]	(5.57)	
Medicare payment per treatment	385.06	388.25	−2.04	265
	[38.86]	[36.63]	(4.58)	
Net income from patient services	19.23	13.99	5.25**	266
	[21.57]	[14.52]	(2.55)	
Net income per treatment	97.74	85.48	12.90	265
	[134.73]	[51.64]	(14.77)	
Panel D. Clinical indicators				
Vascular-access index	0.117	−0.051	0.166*	301
	[0.807]	[0.829]	(0.100)	
Long-term catheter share	12.18	12.14	0.04	301
	[3.71]	[3.59]	(0.45)	
Fistula-use share	62.93	65.21	−2.26***	301
	[6.91]	[6.99]	(0.82)	
Mortality standardized ratio	0.987	0.996	−0.009	301
	[0.115]	[0.142]	(0.014)	
Hospital-admission standardized ratio	0.974	0.970	0.004	301
	[0.121]	[0.125]	(0.015)	
Readmission standardized ratio	0.996	0.998	−0.003	300
	[0.101]	[0.139]	(0.014)	
Dialysis adequacy share	0.969	0.969	0.000	301
	[0.011]	[0.017]	(0.002)	

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Table A2: Additional Baseline Characteristics and Predetermined Controls (continued)

Characteristic	ETC Mean [SD]	Control Mean [SD]	Difference (SE)	Obs.
Panel E. Geographic availability				
Distance to nearest home-capable facility	13.38	13.06	0.36	302
	[20.45]	[19.70]	(2.54)	
Home-capable facilities within 15 miles	5.09	5.93	−0.83	302
	[7.59]	[9.29]	(0.99)	
CKD risk beyond 30 miles	8.29	9.39	−1.05	302
	[13.84]	[19.20]	(1.97)	

Notes: HRR-level differences adjust for randomization regions. Baselines are from 2019, except COVID-19 mortality through September 17, 2020. Monetary totals are millions of 2024 dollars; monetary ratios are dollars per treatment. Patient and treatment totals are thousands. Shares are percentages except adequacy, a fraction. Distances are miles, age is years, and COVID-19 mortality is deaths per 100,000. FTE denotes full-time equivalent. Parentheses report HC2 standard errors. Stars denote the 10, 5, and 1 percent levels (*, **, ***) from 10,000 region-stratified studentized reassignments. The frozen 2019 market-size control totals 6,921 facilities; the reconciled incumbent roster contains 6,971.

Table A3: Population and Outcome Definitions

Measure	Definition
Panel A. Populations and recorded supply	
Form 265 eligible facility	A nonhospital renal facility whose CMS Certification Number (CCN) ends in 2500–2899 or 2900–2999. These Form 265 categories define the reporting scope independently of ownership and report availability.
Incumbent facilities	Eligible facilities operating in 2019, including entrants that year. Membership is fixed; physical exit requires evidence beyond administrative identifier termination.
Original eligible cohort size; N_{hc}	Eligible physical locations entering HRR h in year c. The fixed original roster includes known nonproviders and subsequent exits.
Covered-market outcomes	Annual HRR aggregates of eligible facilities, including later openings, administrative successors, and locations with unresolved physical entry dates. The 2025 extension follows the verified 2024 roster.
Home dialysis–providing facilities; P_{hct}	Cohort facilities with positive recorded Medicare home-dialysis activity in year t. A positive component can establish provision even when complete annual treatment quantity is unresolved.
Providing home dialysis	An indicator for positive recorded Medicare home activity, with zero for known nonproviders and missing for unresolved status. Facility probability effects are percentage points.
Medicare home treatments; q_{it}, Q_{hct}	Facility quantity is Worksheet D, column 4, lines $7 + 8 + 3(9 + 10)$; weekly peritoneal-dialysis entries count as three treatment equivalents. Q_{hct} sums these quantities for the fixed cohort. Patients can contribute multiple treatments.

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Table A3: Population and Outcome Definitions (continued)

Measure	Definition
Mean service per eligible entrant; $\bar{q}$	Medicare home treatments divided by the original entrant count. The accounting exercise takes ratios of jointly standardized quantities and counts, retaining nonproviders and exits in the denominator.
Baseline treatments per providing facility	Medicare home treatments divided by home dialysis–providing facilities in 2019, defined for a positive observed provider count. This ratio describes baseline treatment intensity in Table 1.
Home-service share; provision proportion p_{hct}	Providing facilities divided by the original baseline or cohort roster. The decomposition takes the ratio of standardized provider and entrant counts; empty local cohorts have undefined proportions.
Panel B. Training, provision histories, and accounting	
Training treatments	All-payer training treatments from Worksheet D, column 1, lines 3–6. Reports last 330–400 days, cover at least 90 percent of the calendar year, and start no earlier than entry; follow-up begins in $c+1$. Trainees can contribute multiple treatments; completion is unobserved.
All-payer home and total treatments	Worksheet D, column 1: home treatments sum $7+8+3(9+10)$; total treatments sum lines 1–8 plus $3(9+10)$. Matched quantities exclude separately reported acute-kidney-injury activity.
All-payer home-treatment share	The matched all-payer home-treatment quantity divided by total treatment equivalents, using the preceding Worksheet D definition.
Continuous provision from first full year	Positive provision in every year from $c+1$ through t. A known zero resolves the history to zero; otherwise an unresolved required year leaves it missing. The first observation is one-year status.
Provision in both 2023 and 2024	Positive activity in both years on the original cohort roster, applying the same zero/missing rule as continuous provision. For 2023 entrants, this window includes the potentially partial entry year.
Count and service components	Symmetric allocations in equations (4) and (5). Each pair sums to its standardized linear contrast, in providing facilities or treatment equivalents per HRR.
Panel C. Initial configuration and operator composition	
Home dialysis training capability	Separate early POS indicators for home hemodialysis and home peritoneal-dialysis training and support; these record authorized capability.
Dialysis service capability	Separate early POS indicators for hemodialysis and peritoneal-dialysis services. Near-universal capability may leave insufficient variation for a finite comparison.
Registered nurses	Registered-nurse full-time equivalents (FTEs) in the accepted early POS record. Implausible all-zero personnel blocks remain missing.
Licensed practical nurses	Staff count in the accepted early POS record, retaining the source unit.
Dialysis / technical staff	Separate early FTE fields for dialysis technicians and water-treatment/machine technical staff from the validated POS dictionary.
Dietitians / social workers / other personnel	Three separate early FTE fields from the accepted POS personnel block; invalid blocks retain missing status.

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Table A3: Population and Outcome Definitions (continued)

Measure	Definition
Dialysis stations	Dated, address-verified DFC station counts, supplemented by accepted positive POS counts or dated state capacity evidence. These measure authorized capacity at observation, with opening-day equipment use unobserved.
Big Two / other-chain / independent entrants	Dated operator classification as DaVita/Fresenius, another chain, or independent, with unknown ownership separate. Outcomes are counts per HRR or shares of entrants.
Documented operator experience	Linked pre-policy operator facility counts, home-service shares, and presence in the entrant's HRR, retaining documented histories.
Panel D. Operating inputs	
Total operating expenses	Reported operating expenses aggregated over the eligible HRR population, in 2024 dollars.
Expenses per treatment	Reported operating expenses divided by matched treatment equivalents, in 2024 dollars per treatment.
Clinical / administrative staffing	Occupation-specific employee and contract FTEs from the frozen HCRIS dictionary; these report-period totals differ from early POS snapshots.
Clinical FTE per 100 patients; treatments per FTE	100 times clinical FTEs divided by all-payer patients; and all-payer treatment equivalents divided by clinical FTEs, respectively.
Direct-care compensation	Reported compensation for direct-care labor under the validated HCRIS accounting mapping, aggregated in 2024 dollars.
Cost-report fields for operating inputs	Operating expenses: Worksheet C, line 18, column 2. All-payer treatment denominator: C, line 19, column 1, with the documented quantity reconciliation. Clinical FTEs: S-1, lines 23–26, column 3; administrative FTEs: S-1, lines 29–30, column 3. Direct-care compensation: A, lines 7 and 8, column 8. Stations and patient counts retain their dated registry/report fields.
Panel E. Financial and clinical outcomes	
Medicare payments; payments per treatment	Reported Medicare payment totals and their matched per-treatment ratio, in 2024 dollars. These are facility accounting receipts; total public spending per patient requires additional records.
Net patient-service income	Signed patient-service accounting income, reported as an HRR total or per matched treatment in 2024 dollars. Consolidated company and economic profits remain unobserved.
Mortality / hospitalization / readmission ratios	Three source-defined standardized clinical ratios from public dialysis reports, retaining each event definition, benchmark, and aggregation weights.
Long-term catheter / fistula share	Separate vascular-access shares in the public reports' source-defined patient populations. Effects are percentage points; comparable 2020 catheter observations are unavailable.
Vascular-access index	$\frac{1}{2}(z_{\text{catheter}} - z_{\text{fistula}})$ on joint support, using means and standard deviations from eligible 2019 HRRs. The index uses verified observations; lower values are favorable.
Dialysis adequacy	The public-report fraction meeting the source-defined adequacy criterion, aggregated with the specified source weights and retained on its fractional scale.

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Table A3: Population and Outcome Definitions (continued)

Measure	Definition
Cost-report fields for financial outcomes	Medicare payments and treatment equivalents: Worksheet D, line 11, columns 8 and 4, respectively. Net patient-service revenue and net income: F-1, lines 3 and 18, column 2. Ratios use matched facility records with a positive aggregate denominator; monetary amounts are deflated to 2024 dollars.
Panel F. Geographic availability	
Nearest home-capable facility	Fixed-demand-weighted miles to the nearest eligible facility with documented administrative home-service capability, using a national search across HRR boundaries.
Home-capable facilities within 15 miles	Fixed-demand-weighted number of administratively home-capable facilities within 15 miles, using the same national search.
Home-service desert within 30 miles	Share of the fixed weighted demand population with no administratively home-capable facility within 30 miles. Effects are percentage points; the search crosses HRR boundaries.

Notes: Outcomes retain verified records without statistical imputation; known zeros and unresolved activity remain distinct. Ratios use matched observations and positive denominators. Early configuration is the first accepted dated observation within 180 days of entry. Administrative capability differs from recorded treatment activity. Section 3.2 defines windows and samples; Table A1 identifies sources.

Table A4 describes the dispersion and prevalence of zeros in the main outcomes. Table A5 reports service coverage within the fixed main HRRs and training coverage by assignment and year. Of the 149, 108, 80, and 63 classified entrants in 2021–2024, the respective main samples retain 135, 101, 70, and 56: 362 of the 400 eligible locations. The incumbent sample retains 6,970 of 6,971 locations. HRRs are retained according to the availability of observed components; requiring complete facility histories is a separate robustness restriction. Empty local cohorts contribute zero, partially observed populations contribute known activity, and entirely unresolved nonempty populations remain unavailable. Training uses a fixed subset with observed aggregate outcomes in every eligible year and begins after entry under the full-report rule, leaving the 2024 cohort without a main-window observation. The coverage rates measure the share of facility records with resolved outcomes, leaving uncertainty about whether missingness is independent of assignment.

The entry review distinguishes corroborated new openings, administratively classified openings without identified prior operations, and records with evidence of an existing site. New York's January 2022 licensing review describes the acquisition of existing Faxton, Herkimer, and St. Luke's clinics at the addresses later associated with CCNs 332852, 332854, and 332857. They are excluded from the 2023 new-location cohort. In Florida, CCN 682742 shares an address and suite with the earlier CCN 682686; original reports and enrollment records support exclusion from the 2022 new-location cohort but leave the intervening continuity of service

unresolved.[50] Their observed service remains in the covered-market category outside the classified new-location cohorts. Remaining alias and same-site ambiguities limit interpretation of administrative facility counts as distinct physical capacity. Previously excluded relocation and pre-existing-site cases remain excluded; exit zeros require evidence of physical closure.

Source selection preserves report identity and period before aggregating values. For a given reporting interval, accepted revisions supersede the earlier filing. Calendar-year service totals use documented report intervals and exact reconciliations of nonoverlapping periods where available; an unresolved interval remains missing. The training outcome instead uses the qualifying report total attributed to its reference year under the 330–400-day and 90-percent-coverage rule. Input fields retain their accepted annual report mappings, including partial entry-year periods. Recoveries require compatible identities, explicit worksheet evidence, and the field's period rule. A blank cell is treated as a source-coded zero only when companion cells and worksheet arithmetic establish that coding; report absence is never sufficient. The archive retains source-quality flags for unresolved staffing units, period conflicts, and identities, without inferring decimal corrections from neighboring facilities. These conventions improve documented coverage but cannot eliminate source errors or assignment-related reporting.

Table A4: Distributions of Main Supply Outcomes

Population / outcome	HRRs	Mean range	Zero share range	Maximum
2020 entrants: Home dialysis–providing facilities	290	0.17–0.26	81.4–86.6	7.0
2020 entrants: Medicare home treatments	290	46.73–317.16	81.7–87.2	12,024.0
2021 entrants: Home dialysis–providing facilities	290	0.18–0.24	83.4–85.9	4.0
2021 entrants: Medicare home treatments	290	63.76–339.68	83.4–86.2	7,812.0
2022 entrants: Home dialysis–providing facilities	296	0.16–0.20	86.5–87.8	4.0
2022 entrants: Medicare home treatments	296	39.59–175.63	86.8–87.8	5,889.0
2023 entrants: Home dialysis–providing facilities	293	0.11–0.14	90.4–91.5	4.0
2023 entrants: Medicare home treatments	293	29.65–112.46	90.4–91.5	4,044.0

Continued on next page

[50]The New York evidence is in the January 27, 2022 Public Health and Health Planning Council exhibits, PDF pages 82–83 and 86. The Florida evidence includes original cost reports and NPPES records 1184141350 and 1558095513. These records do not resolve the August–September 2022 interval, so no continuous facility history or additional treatment quantity is imputed.

Table A4: Distributions of Main Supply Outcomes (continued)

Population / outcome	HRRs	Mean range	Zero share range	Maximum
2024 entrants: Home dialysis–providing facilities	295	0.09–0.09	91.2–91.2	2.0
2024 entrants: Medicare home treatments	295	30.98–30.98	91.2–91.2	2,325.0
Incumbents: Home dialysis–providing facilities	301	9.55–10.21	1.7–2.3	120.0
Incumbents: Medicare home treatments	301	12,117.09–17,683.17	1.7–3.0	164,118.0
Covered market: Home dialysis–providing facilities	301	9.99–10.78	0.7–1.7	123.0
Covered market: Medicare home treatments	301	13,311.79–17,766.90	1.0–2.0	164,229.0

Notes: Ranges summarize annual cross-sections on fixed HRRs from 2019 or cohort entry through 2024. Zero shares are percentages, including empty cohorts; means and maxima retain the stated outcome units.

Table A5: Facility Reporting Coverage

Population / measure	2019	2020	2021	2022	2023	2024
Panel A. Main supply records within fixed HRRs						
Incumbents						
Fixed HRRs	301	301	301	301	301	301
Required facility records	6,970	6,970	6,970	6,970	6,970	6,970
Provision status known (percent)	96.9	97.0	97.0	97.0	96.9	96.9
Treatment quantity known (percent)	95.7	95.7	95.2	95.9	95.8	96.1
Covered market						
Fixed HRRs	301	301	301	301	301	301
Required facility records	6,970	7,180	7,348	7,481	7,570	7,648
Provision status known (percent)	96.9	96.8	96.8	96.9	96.7	96.8
Treatment quantity known (percent)	95.7	95.5	95.1	95.7	95.5	95.8
2020 entrants						
Fixed HRRs	—	290	290	290	290	290
Required facility records	—	177	177	177	177	177
Provision status known (percent)	—	93.8	97.7	98.9	98.3	98.3
Treatment quantity known (percent)	—	91.5	95.5	97.2	95.5	97.2
2021 entrants						
Fixed HRRs	—	—	290	290	290	290
Required facility records	—	—	135	135	135	135

Continued on next page

Table A5: Facility Reporting Coverage (continued)

Population / measure	2019	2020	2021	2022	2023	2024
Provision status known (percent)	—	—	96.3	98.5	99.3	99.3
Treatment quantity known (percent)	—	—	94.8	96.3	99.3	97.8
2022 entrants						
Fixed HRRs	—	—	—	296	296	296
Required facility records	—	—	—	101	101	101
Provision status known (percent)	—	—	—	94.1	97.0	98.0
Treatment quantity known (percent)	—	—	—	94.1	94.1	95.0
2023 entrants						
Fixed HRRs	—	—	—	—	293	293
Required facility records	—	—	—	—	70	70
Provision status known (percent)	—	—	—	—	97.1	98.6
Treatment quantity known (percent)	—	—	—	—	97.1	97.1
2024 entrants						
Fixed HRRs	—	—	—	—	—	295
Required facility records	—	—	—	—	—	56
Provision status known (percent)	—	—	—	—	—	96.4
Treatment quantity known (percent)	—	—	—	—	—	96.4
Panel B. Training reports: known records / original facility roster						
2020 entrants						
Control	—	—	129 / 135	130 / 135	128 / 135	128 / 135
ETC	—	—	42 / 42	42 / 42	41 / 42	42 / 42
2021 entrants						
Control	—	—	—	80 / 83	80 / 83	82 / 83
ETC	—	—	—	52 / 52	52 / 52	52 / 52
2022 entrants						
Control	—	—	—	—	65 / 67	65 / 67
ETC	—	—	—	—	33 / 34	32 / 34
2023 entrants						
Control	—	—	—	—	—	45 / 48
ETC	—	—	—	—	—	21 / 22

Notes: Panel A divides known outcomes by all required facility records in the fixed main HRRs. Known nonproviders and documented exits are resolved zeros; unknown contributions remain unresolved. Panel B reports known training records and original facility counts separately by assignment. Training requires a 330–400-day report covering at least 90 percent of the calendar year; its regressions further restrict HRRs to a fixed observed subset. High coverage alone does not establish assignment-independent reporting. Dashes indicate years outside the relevant window.

B Supplementary Estimation and Inference

B.1 Facility Comparisons and Initial Configuration

For facility i in cohort c, the training and provision-history comparisons use

$$y_{ict} = \lambda_{s(h(i)),ct} + \beta_{ct} T_{h(i)} + \rho_{ct} Y_{h(i),2019} + \boldsymbol{\gamma}'_{ct} \mathbf{X}_{h(i)} + \epsilon_{ict}. \tag{A1}$$

Training uses the 2019 market home-treatment baseline, and provision histories use the home-provider baseline. The eight controls and assignment are measured at the facility's HRR. Comparisons weight observed facilities equally within the corresponding fixed aggregate geography, clustering standard errors and reassigning treatment at the HRR level. Because entry and reporting can respond to assignment, these coefficients describe the facilities observed under each assignment, including differences in entrant composition.

Initial configuration uses the first accepted observation z_{ic} within 180 days of entry:

$$z_{ic} = \lambda_{s(h(i)),c} + \delta_c T_{h(i)} + \rho_c Y_{h(i),2019} + \boldsymbol{\gamma}'_c \mathbf{X}_{h(i)} + v_{ic}. \tag{A2}$$

Home-training capabilities use the 2019 home-provider baseline; other configuration measures use the 2019 eligible-facility count. The regressions retain dated records within the approved cohort HRRs. Selection into entry and the availability of early records also qualify these conditional comparisons.

B.2 Randomization and Coefficient Averages

Randomization inference preserves the original regional treatment totals over all 302 HRRs, even when a regression uses a subset. The Midwest contains 25 treated and 59 control HRRs, the Northeast 13 and 30, the South 35 and 80, and the West 18 and 42. The observed outcomes, covariates, and sample masks are held fixed. Validity for this selected sample therefore requires its membership and measurement to be invariant to reassignment under the null; fixing the observed mask in the computation does not establish that condition.

For each specification, I first fit the nuisance model imposing zero treatment coefficients. Let $V(z)$ contain the assignment regressors under reassignment z, residualized against the nuisance design, and let r denote the null residual vector. In OLS, residualization is unweighted. In PPML, the design is weighted by the square root of the null fitted mean and r is the corresponding Pearson residual; HRR effects are included in the projection where applicable. The score-based coefficient vector is $\widetilde{\boldsymbol{\tau}}(z) = [V(z)'V(z)]^{+}V(z)'r$, where $+$ denotes the generalized inverse. Its HRR influence vector is $\psi_h(z) = [V(z)'V(z)]^{+} \sum_{i \in h} V_i(z) r_i / \sqrt{1 - \ell_i}$,

with ℓ_i the null-model leverage, including absorbed HRR effects. For contrast weights w, the statistic is

$$S_w(z) = \frac{w'\widetilde{\boldsymbol{\tau}}(z)}{\{\sum_h [w'\psi_h(z)]^2\}^{1/2}}. \tag{A3}$$

Annual contrasts select one coefficient; average contrasts weight the included annual coefficients equally. Summing the HRR influence vectors before squaring retains cross-year covariance. The two-sided comparison ranks $|S_w|$ under the observed assignment against 10,000 reassignments, with the observed assignment included in the Monte Carlo correction. The tabled point estimates and HC2 standard errors come from the unrestricted regressions, while the score statistic uses the restricted nuisance fit. Undefined statistics are reported as unavailable, without treating them as evidence of insignificance.

The reference null is no assignment effect on any represented unit and year. An average-score statistic is sensitive to a signed average departure, rather than being an omnibus test for every possible annual pattern. Rejection supplies evidence against this sharp null; validity for a weak zero-average null with heterogeneous effects requires additional arguments (Wu and Ding 2021). Ownership and operator-composition comparisons use fully re-estimated, studentized linear statistics under the same assignments. Significance refers to individual comparisons without a simultaneous adjustment across the reported outcomes. Conventional HC2 intervals characterize uncertainty in coefficient averages and can differ from the RI assessment.

C Supplementary Supply and Entrant Evidence

C.1 The 2020 Transition Cohort

Table A6 collects the 2020 cohort's supply, training, continued provision, operating inputs, initial configuration, and operator characteristics. Its original roster contains 189 eligible facilities, and its core panel retains 290 HRRs. Core service and input averages include entry; training and provision histories follow the windows in Section 3.2.

Table A6: The 2020 Transition Cohort: Supplementary Outcomes

Outcome / unit	ETC effect	Standard error	HRRs
Panel A. Home service, training, and continued provision			
Providing facilities per HRR	−0.048	0.059	290

Continued on next page

Table A6: The 2020 Transition Cohort: Supplementary Outcomes (continued)

Outcome / unit	ETC effect	Standard error	HRRs
Medicare home treatments per HRR	−86.13	82.25	290
Training treatments per HRR	−33.03	28.70	287
Training treatments per observed facility	−41.33	45.67	93
Provision in every full follow-up year (facilities per HRR)	−0.042	0.058	290
Provision in every full follow-up year (pp)	0.93	8.35	96
Provision in both 2023 and 2024 (facilities per HRR)	−0.062	0.069	290
Provision in both 2023 and 2024 (pp)	−0.18	8.44	96
Panel B. Operating inputs			
Operating expenses (million dollars)	−0.207	0.241	278
Dialysis stations	0.075	1.192	279
Clinical FTEs	−0.652	0.737	276
Administrative FTEs	−0.126	0.175	276
Direct-care compensation (million dollars)	−0.061	0.069	278
Expenses per treatment (dollars)	35.05	123.11	81
Clinical FTEs per 100 patients	0.08	0.88	79
Treatments per clinical FTE	−141.78	96.53	80
Panel C. Initial configuration			
Home hemodialysis training and support (pp)	1.71	6.16	88
Home peritoneal-dialysis training and support (pp)	−10.83	7.74	88
Registered nurses (FTEs)	0.54	0.70	88
Dialysis stations	−0.85	1.28	68
Hemodialysis service (pp)	−4.75	8.52	88
Peritoneal-dialysis service (pp)	4.87	4.55	88
Licensed practical nurses (staff count)	−0.01	0.06	88
Dialysis technicians (FTEs)	0.79	0.62	88
Water/machine technicians (FTEs)	0.09	0.09	88
Dietitians (FTEs)	−0.21	0.17	88
Social workers (FTEs)	−0.17	0.16	88
Other staff (FTEs)	−0.27	0.20	88
Panel D. Entrant ownership counts per HRR			
DaVita and Fresenius	−0.128	0.103	290
Other chains	0.000	0.048	290
Independent facilities	−0.033	0.035	290
Unknown ownership	0.001	0.020	290
Panel E. Ownership shares (pp)			
DaVita and Fresenius	−4.667	9.804	290
Other chains	5.649	8.118	290
Independent facilities	−1.860	5.808	290
Unknown ownership	0.878	3.662	290
Panel F. Documented operator experience			
Operator facilities in 2018	−202.805	208.757	290

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Table A6: The 2020 Transition Cohort: Supplementary Outcomes (continued)

Outcome / unit	ETC effect	Standard error	HRRs
Operator home hemodialysis share in 2018 (pp)	0.859	1.918	290
Operator home peritoneal-dialysis share in 2018 (pp)	2.069	1.384	290
Local operator facilities in 2019	−13.820*	7.908	290

Notes: Core service and input averages cover 2020–2024; training and growing provision histories cover 2021–2024. The common provision outcome requires activity in both 2023 and 2024. Averages weight years equally. Configuration uses the first accepted record within 180 days of entry. Ownership and experience use standardized HRR counts or ratios with region effects and eight controls; experience retains documented histories. Facility comparisons condition on entry and reporting. Input totals are per HRR; amounts are 2024 dollars and ratios use positive matched denominators. pp denotes percentage points; FTE denotes full-time equivalent. Standard errors are HRR-clustered HC2. Stars follow the individual regional randomization convention (*, **, ***: 10, 5, 1 percent); ownership and experience statistics are fully re-estimated.

C.2 Annual Service Estimates

Table A7 reports the annual coefficients underlying the cohort paths in Figure 2 and the incumbent and covered-market paths in Figure 4. Provision and treatments share the same fixed HRRs within each population; the incumbent and market panels retain 301 HRRs. The table includes the 2020 comparisons and reports each annual estimate once. Cohort averages appear in Table 2, incumbent and market averages in Table 4, and the transition-cohort average in Table A6.

Table A7: Annual Home Dialysis Supply by Population

Year	Providing facilities	Medicare home treatments	Control mean: facilities	Control mean: treatments	HRRs	HRR-years
Panel A. 2020 entrants						
2020	−0.050	4.28	0.188	48.30	290	290
	(0.056)	(32.86)				
2021	−0.047	−27.05	0.252	177.87	290	290
	(0.060)	(70.30)				
2022	−0.069	−66.80	0.267	271.71	290	290
	(0.063)	(93.63)				
2023	−0.024	−162.53	0.272	339.19	290	290
	(0.068)	(117.92)				
2024	−0.049	−178.55	0.267	363.93	290	290
	(0.071)	(134.98)				
Panel B. 2021 entrants						
2021	0.087	55.23	0.156	46.16	290	290
	(0.062)	(36.90)				
2022	0.146*	193.86*	0.180	134.62	290	290
	(0.082)	(101.82)				
2023	0.140*	220.57*	0.185	195.34	290	290
	(0.078)	(125.72)				

Continued on next page

Table A7: Annual Home Dialysis Supply by Population (continued)

Year	Providing facilities	Medicare home treatments	Control mean: facilities	Control mean: treatments	HRRs	HRR-years
2024	0.127	262.37	0.205	257.63	290	290
	(0.078)	(158.16)				
Panel C. 2022 entrants						
2022	0.167***	50.73	0.112	24.15	296	296
	(0.062)	(40.66)				
2023	0.118*	97.60	0.165	93.66	296	296
	(0.067)	(72.52)				
2024	0.116*	110.03	0.165	143.62	296	296
	(0.067)	(96.49)				
Panel D. 2023 entrants						
2023	−0.070*	−32.87	0.132	37.12	293	293
	(0.039)	(19.28)				
2024	−0.033	−148.60***	0.151	149.42	293	293
	(0.053)	(50.54)				
Panel E. 2024 entrants						
2024	−0.015	−2.71	0.092	32.06	295	295
	(0.039)	(17.23)				
Panel F. Incumbents						
2020	0.055	14.95	10.286	17,315.19	301	301
	(0.117)	(520.48)				
2021	−0.004	126.07	10.386	15,072.57	301	301
	(0.163)	(540.54)				
2022	0.216	143.90	10.148	13,615.65	301	301
	(0.193)	(692.05)				
2023	0.340*	−348.50	9.881	12,666.93	301	301
	(0.223)	(688.72)				
2024	0.358	−463.96	9.638	11,991.46	301	301
	(0.266)	(813.58)				
Panel G. Covered market						
2020	−0.025	−38.42	10.510	17,416.62	301	301
	(0.129)	(518.17)				
2021	0.013	152.83	10.881	15,447.29	301	301
	(0.196)	(485.37)				
2022	0.358	540.61	10.862	14,277.34	301	301
	(0.245)	(614.19)				
2023	0.439	−393.67	10.810	13,559.32	301	301
	(0.289)	(685.96)				
2024	0.384	−593.87	10.748	13,205.62	301	301
	(0.326)	(797.12)				

Notes: Annual linear estimates follow equations (2) and (1). Cohort follow-up starts at entry; incumbent and covered-market effects are relative to 2019. Provider counts and Medicare home-treatment equivalents are per HRR. Both outcomes use the same fixed HRRs within each population. Totals retain verified contributions from partially observed local rosters; unknown activity is not imputed. Parentheses report HRR-clustered HC2 standard errors. Stars denote the 10, 5, and 1 percent levels (*, **, ***) in individual tests using 10,000 region-stratified reassignments under the sharp no-effect null.

C.3 Accounting Uncertainty and Denominator Diagnostics

Table A8 compares bootstrap and Delta standard errors for the decomposition in Section 5.2. Each draw resamples the same HRRs across equations and years to preserve covariance, and all 2,000 finite draws enter the uncertainty estimates. Economic admissibility additionally requires positive adjusted cohort counts, nonnegative service quantities, and provider counts bounded by original cohort size in both assignment states and every included year. For the 2023 treatment allocation, 781 draws contain a negative fitted quantity in at least one year, limiting a literal count-and-rate interpretation of those draws.

Table A8: Accounting Uncertainty and Denominator Diagnostics

Cohort / outcome	Count bootstrap SE	Rate bootstrap SE	Count Delta SE	Rate Delta SE	Admissible draws
2021 / providers	0.071	0.050	0.071	0.049	2,000
2021 / treatments	72.891	81.849	71.975	79.137	2,000
2022 / providers	0.045	0.042	0.044	0.040	1,990
2022 / treatments	28.579	54.680	28.124	52.329	1,999
2023 / providers	0.033	0.029	0.032	0.027	1,971
2023 / treatments	16.051	32.322	15.449	30.737	1,219
2024 / providers	0.025	0.029	0.024	0.028	1,996
2024 / treatments	9.716	17.183	8.439	16.043	1,981

Notes: Standard errors use all 2,000 finite HRR bootstrap draws. Admissibility requires positive adjusted entrant counts and nonnegative service in both arms, with provider counts bounded by entrant counts in every averaged year. Delta-method calculations retain covariance across equations and years on fixed samples.

C.4 Continued Provision

Table A9 compares provision in both 2023 and 2024 and provision in every year from the first full follow-up year onward.[51] The 2021 and 2022 cohorts contribute an additional 0.140 and 0.120 facilities per HRR providing in both years, respectively; both increases are statistically significant. Their growing-history count responses are similar, while conditional facility probabilities remain imprecise. For the 2023 cohort, the common-window estimates are -0.067 facilities per HRR and -39.80 percentage points at the facility level; only the latter is statistically significant. The early cohorts' continuing supply contribution therefore coincides with a larger set of active locations, while these comparisons leave individual persistence and selection into entry difficult to separate.

[51] For 2023 entrants, the common window includes entry, and the first-full-year measure covers 2024 alone. The 2024 cohort has no corresponding follow-up. Both measures retain the original roster. A known zero resolves a history to zero; otherwise an unresolved required year leaves it missing.

Table A9: Continued Home Dialysis Provision

Outcome / level	2021	2022	2023	2024
Provision in every full follow-up year / HRR count				
ETC effect	0.138*	0.119*	−0.033	—
	(0.080)	(0.067)	(0.053)	—
Control mean	0.18	0.16	0.15	—
HRR-years	870	592	293	—
HRRs	290	296	293	—
Provision in both 2023 and 2024 / HRR count				
ETC effect	0.140*	0.120*	−0.067	—
	(0.078)	(0.067)	(0.039)	—
Control mean	0.19	0.16	0.13	—
HRR-years	290	296	293	—
HRRs	290	296	293	—
Provision in every full follow-up year / facility probability				
ETC effect	−3.47	17.55	−19.30	—
	(11.13)	(12.16)	(15.59)	—
Control mean	45.31	51.94	65.96	—
Facility-years	401	197	69	—
HRRs	73	66	44	—
Provision in both 2023 and 2024 / facility probability				
ETC effect	−2.34	18.29	−39.80***	—
	(10.59)	(12.08)	(11.88)	—
Control mean	46.34	50.77	54.17	—
Facility-years	134	99	70	—
HRRs	73	66	44	—

Notes: Linear estimates follow equations (2) and (A1). Continuous provision requires positive activity from the first full year after entry through each follow-up year; annual effects are averaged equally through 2024. The common-window outcome requires provision in 2023 and 2024. A known zero resolves a history to zero; otherwise incomplete histories remain missing. Facility effects are percentage points, conditional on observed entrants; control means are percentages. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. Dashes denote unavailable outcomes.

C.5 Initial Configuration and Entrant Composition

Table A10 describes the initial configuration and operator composition of the 2021–2024 cohorts.[52] Configuration uses dated first-180-day records and equation (A2). Ownership counts, ownership shares, and documented operator experience use standardized HRR totals or ratios with region effects and the eight controls, without a separate outcome baseline. These comparisons describe who enters and the resources recorded near entry; latent capabilities and subsequent investment remain unobserved. The 2020 comparisons are collected in Table A6.

[52]The 2023 station sample is insufficient for the specified regression. Unknown operator histories remain missing.

Table A10: Initial Configuration and Observable Entrant Composition

Outcome / statistic	2021	2022	2023	2024
Panel A. Dated configuration within 180 days of entry				
Home hemodialysis training and support (pp)	−3.35	2.81	−10.94	5.21
	(8.99)	(9.16)	(16.97)	(16.45)
Control mean	12.50	15.00	25.00	25.71
Facilities / HRRs	118 / 65	90 / 61	52 / 35	49 / 40
Home peritoneal-dialysis training and support (pp)	−3.64	−1.21	−2.36	−3.06
	(10.01)	(13.26)	(21.19)	(19.72)
Control mean	31.94	50.00	47.22	37.14
Facilities / HRRs	118 / 65	90 / 61	52 / 35	49 / 40
Registered nurses (FTEs)	1.59	0.11	−1.44	−0.13
	(0.84)	(0.61)	(1.48)	(0.30)
Control mean	2.32	2.11	2.17	1.92
Facilities / HRRs	115 / 64	89 / 60	52 / 35	49 / 40
Dialysis stations	4.33	−5.84**	—	3.23
	(3.46)	(2.34)	—	(3.45)
Control mean	13.11	13.78	12.83	12.41
Facilities / HRRs	41 / 30	40 / 33	6 / 3	33 / 28
Hemodialysis service (pp)	1.73	−4.12	12.46	−9.16
	(8.42)	(11.20)	(15.87)	(11.56)
Control mean	76.39	71.67	72.22	82.86
Facilities / HRRs	118 / 65	90 / 61	52 / 35	49 / 40
Peritoneal-dialysis service (pp)	−1.29	3.03	−25.08**	−14.14
	(3.88)	(11.76)	(9.54)	(9.96)
Control mean	4.17	10.00	11.11	11.43
Facilities / HRRs	118 / 65	90 / 61	52 / 35	49 / 40
Licensed practical nurses (staff count)	0.11	−0.44*	0.00	0.11
	(0.08)	(0.34)	(0.06)	(0.10)
Control mean	0.15	0.27	0.06	0.13
Facilities / HRRs	118 / 65	90 / 61	52 / 35	49 / 40
Dialysis technicians (FTEs)	0.51	−0.41	−0.12	0.06
	(0.39)	(0.43)	(0.31)	(0.40)
Control mean	1.00	1.36	1.04	1.26
Facilities / HRRs	118 / 65	90 / 61	52 / 35	49 / 40
Water/machine technicians (FTEs)	−0.12	0.17	0.06	0.06
	(0.12)	(0.19)	(0.29)	(0.19)
Control mean	0.66	0.56	0.92	0.59
Facilities / HRRs	118 / 65	90 / 61	52 / 35	49 / 40
Dietitians (FTEs)	−0.14	0.01	0.13	0.13
	(0.10)	(0.10)	(0.22)	(0.15)
Control mean	0.81	0.58	0.87	0.59
Facilities / HRRs	118 / 65	90 / 61	51 / 35	49 / 40

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Table A10: Initial Configuration and Observable Entrant Composition (continued)

Outcome / statistic	2021	2022	2023	2024
Social workers (FTEs)	−0.23	−0.55	0.12	0.14
	(0.13)	(0.41)	(0.22)	(0.14)
Control mean	0.84	0.91	0.87	0.58
Facilities / HRRs	118 / 65	90 / 61	52 / 35	49 / 40
Other staff (FTEs)	0.27	0.60	1.06*	−0.33
	(0.19)	(0.57)	(0.48)	(0.23)
Control mean	0.26	0.43	0.54	0.42
Facilities / HRRs	118 / 65	90 / 61	52 / 35	49 / 40
Panel B. Entrant counts per HRR				
DaVita and Fresenius	0.066	0.044	−0.067	0.054
	(0.089)	(0.058)	(0.044)	(0.039)
Adjusted control mean	0.20	0.18	0.13	0.07
Other chains	0.203	0.032	−0.003	−0.030
	(0.122)	(0.037)	(0.033)	(0.020)
Adjusted control mean	0.09	0.06	0.05	0.04
Independent facilities	−0.008	0.013	0.062	−0.041
	(0.034)	(0.035)	(0.044)	(0.028)
Adjusted control mean	0.08	0.07	0.05	0.08
Unknown ownership	−0.010	−0.007	0.014	−0.010
	(0.007)	(0.008)	(0.016)	(0.008)
Adjusted control mean	0.01	0.01	0.01	0.01
Panel C. Ownership shares (pp)				
DaVita and Fresenius	−10.157	−0.229	−29.081**	36.739**
	(12.263)	(9.832)	(12.757)	(16.245)
Adjusted control mean	52.26	55.53	56.03	34.76
Other chains	22.409	3.783	−1.877	−14.468
	(12.719)	(8.879)	(11.842)	(9.655)
Adjusted control mean	23.52	20.44	20.57	18.18
Independent facilities	−9.295	−1.142	25.105*	−17.185
	(6.403)	(7.849)	(13.835)	(13.266)
Adjusted control mean	20.80	21.20	19.47	42.12
Unknown ownership	−2.958	−2.411	5.852	−5.086
	(1.761)	(2.519)	(6.682)	(4.324)
Adjusted control mean	3.42	2.83	3.93	4.94
Panel D. Documented operator experience				
Operator facilities in 2018	−398.330	−262.509	−363.180	788.285***
	(311.075)	(211.903)	(357.127)	(242.567)
Adjusted control mean	1,686.57	1,941.92	1,835.07	1,558.22
Operator home hemodialysis share in 2018 (pp)	−1.516	−2.208	10.749	3.052
	(2.058)	(1.777)	(10.159)	(3.627)
Adjusted control mean	20.58	20.90	21.45	22.57
Operator home peritoneal-dialysis share in 2018 (pp)	2.921*	−0.210	7.095	−6.279***
	(1.741)	(2.471)	(6.194)	(2.245)
Adjusted control mean	43.90	43.01	43.92	47.90

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Table A10: Initial Configuration and Observable Entrant Composition (continued)

Outcome / statistic	2021	2022	2023	2024
Local operator facilities in 2019	−5.588	−9.328*	−13.561	8.604
	(4.070)	(5.062)	(11.907)	(5.133)
Adjusted control mean	18.04	24.01	22.34	10.48
Common support for Panels B–D				
Original entrants	135	101	70	56
HRRs	290	296	293	295

Notes: Panel A follows equation (A2); binary outcomes are percentage points (pp), and staffing retains source units. FTE denotes full-time equivalent. Panels B–D use standardized HRR counts or ratios with region effects and eight controls, without an outcome baseline. Original entrants count facilities; experience averages retain documented histories. The 2023 station comparison lacks sufficient unsaturated support. Parentheses report HRR-clustered HC2 standard errors. Stars follow individual regional randomization tests (*, **, ***: 10, 5, 1 percent), with fully re-estimated statistics in Panels B–D.

C.6 Market Accounting

Table A11 connects the populations using a common linear design. The outcome identity is

$$Y_{ht}^{\text{market}} = Y_{ht}^{I2019} + \sum_{c=2020}^{2024} Y_{ht}^{c} + Y_{ht}^{\text{other}}. \tag{A4}$$

Identical regressors, observations, and weights make the component coefficients add to the market coefficient. The shared samples retain 271 HRRs for providers and 260 for treatments. Pre-entry contributions are structural zeros. The residual contains verified activity from facilities outside the fixed incumbent and classified entry rosters, including the four existing-site records removed from entry. Across all 302 HRRs, the category has 21, 40, 65, 74, and 89 facility records in 2020–2024; the treatment-accounting sample contains 18, 34, 48, 54, and 66. Earlier certification dates in these histories do not by themselves resolve physical entry. On the common treatment sample, the residual contributes −249.2 of the market point estimate of −267.5 equivalents per HRR. Its size makes unresolved entry histories material to interpreting the market decomposition. The exercise establishes an additive identity; it cannot trace patient transfers or attribute the residual to a behavioral mechanism.

Table A11: Additive Market Accounting: Average Supply Contributions

Population component	ETC effect	Control mean	HRR-years	HRRs
Panel A. Home dialysis–providing facilities				
Covered market	0.238	10.005	1,084	271
	(0.248)			

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Table A11: Additive Market Accounting: Average Supply Contributions (continued)

Population component	ETC effect	Control mean	HRR-years	HRRs
Incumbents	0.225	9.228	1,084	271
	(0.199)			
2020 entrants	−0.069	0.266	1,084	271
	(0.075)			
2021 entrants	0.081	0.195	1,084	271
	(0.085)			
2022 entrants	0.084	0.111	1,084	271
	(0.057)			
2023 entrants	−0.026	0.074	1,084	271
	(0.030)			
2024 entrants	0.005	0.020	1,084	271
	(0.010)			
Other observed facilities (residual)	−0.063*	0.112	1,084	271
	(0.038)			
Panel B. Medicare home treatments				
Covered market	−267.5	13,294.8	1,040	260
	(643.3)			
Incumbents	−91.1	12,516.9	1,040	260
	(646.1)			
2020 entrants	−79.0	283.3	1,040	260
	(113.4)			
2021 entrants	126.5	164.5	1,040	260
	(111.6)			
2022 entrants	69.3	65.8	1,040	260
	(64.5)			
2023 entrants	−45.7**	48.4	1,040	260
	(21.3)			
2024 entrants	1.6	7.2	1,040	260
	(5.2)			
Other observed facilities (residual)	−249.2***	208.7	1,040	260
	(105.5)			

Notes: Equal-weight 2021–2024 averages follow equation (A4). Components share observations, weights, and controls, so their coefficients sum to the market effect. Pre-entry values are zero. The residual retains verified activity outside the fixed incumbent and classified entry rosters. Provider counts and Medicare home-treatment equivalents are per HRR. Parentheses report HRR-clustered HC2 standard errors. Stars denote the 10, 5, and 1 percent levels (*, **, ***) in individual tests using 10,000 region-stratified reassignments under the sharp no-effect null.

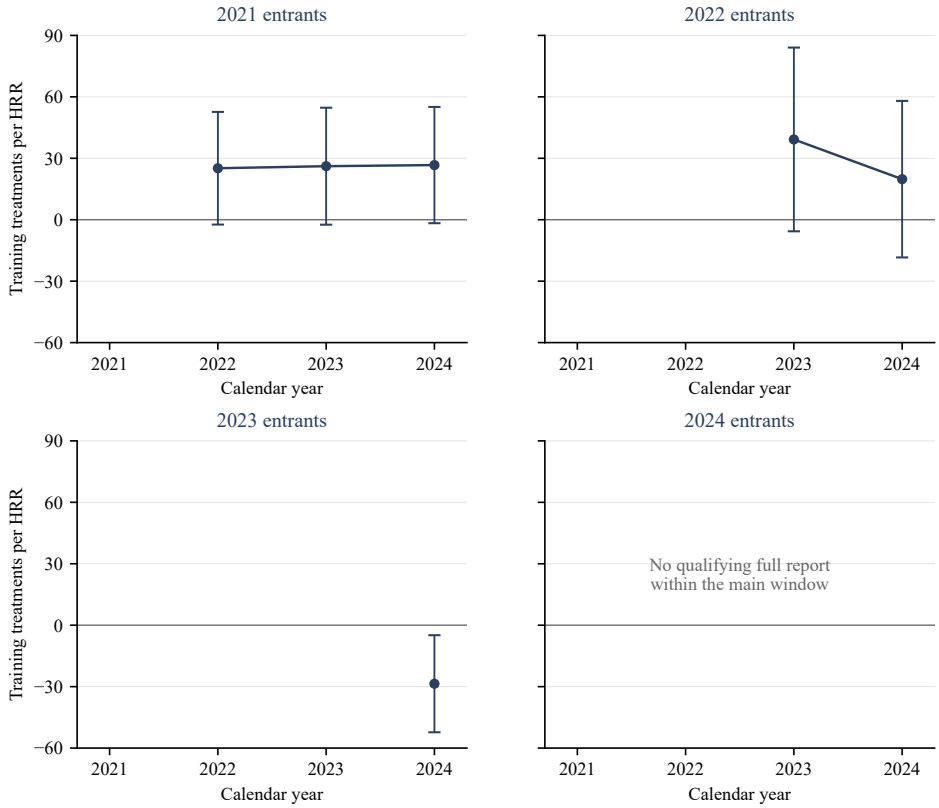
D Training, Inputs, and Other Market Outcomes

D.1 Training Activity

Figure A2 displays the annual HRR training paths discussed in Section 6.1, with positive early-cohort responses and a negative 2023 response. Table A12 reports

annual estimates at the HRR and conditional facility levels and the facility averages; HRR averages appear in Tables 5 and A6. The 2021 HRR average is 25.99 additional training treatments and is statistically significant; its facility average is 21.28 and is imprecise. For 2023 entrants, the respective declines of 28.58 and 139.65 are statistically significant. The evidence for a larger regional training contribution is therefore stronger than that for more training per early entrant. Table A5 reports the underlying coverage by assignment and year.

Figure A2: Reported training activity by entry cohort



Notes: Annual all-payer training treatments per HRR follow equation (2), using the 2019 home-treatment baseline. Fixed cohorts and outcome-specific HRRs cover qualifying full reports after entry through 2024. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Table A12: Annual Reported Training Activity

Cohort / year	HRR effect	HRR control mean	Facility effect	Facility control mean	Facilities / HRRs
Panel A. 2020 entrants					
2021	−41.42	55.06	−52.75	86.03	169 / 93
	(35.46)		(55.49)		

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Table A12: Annual Reported Training Activity (continued)

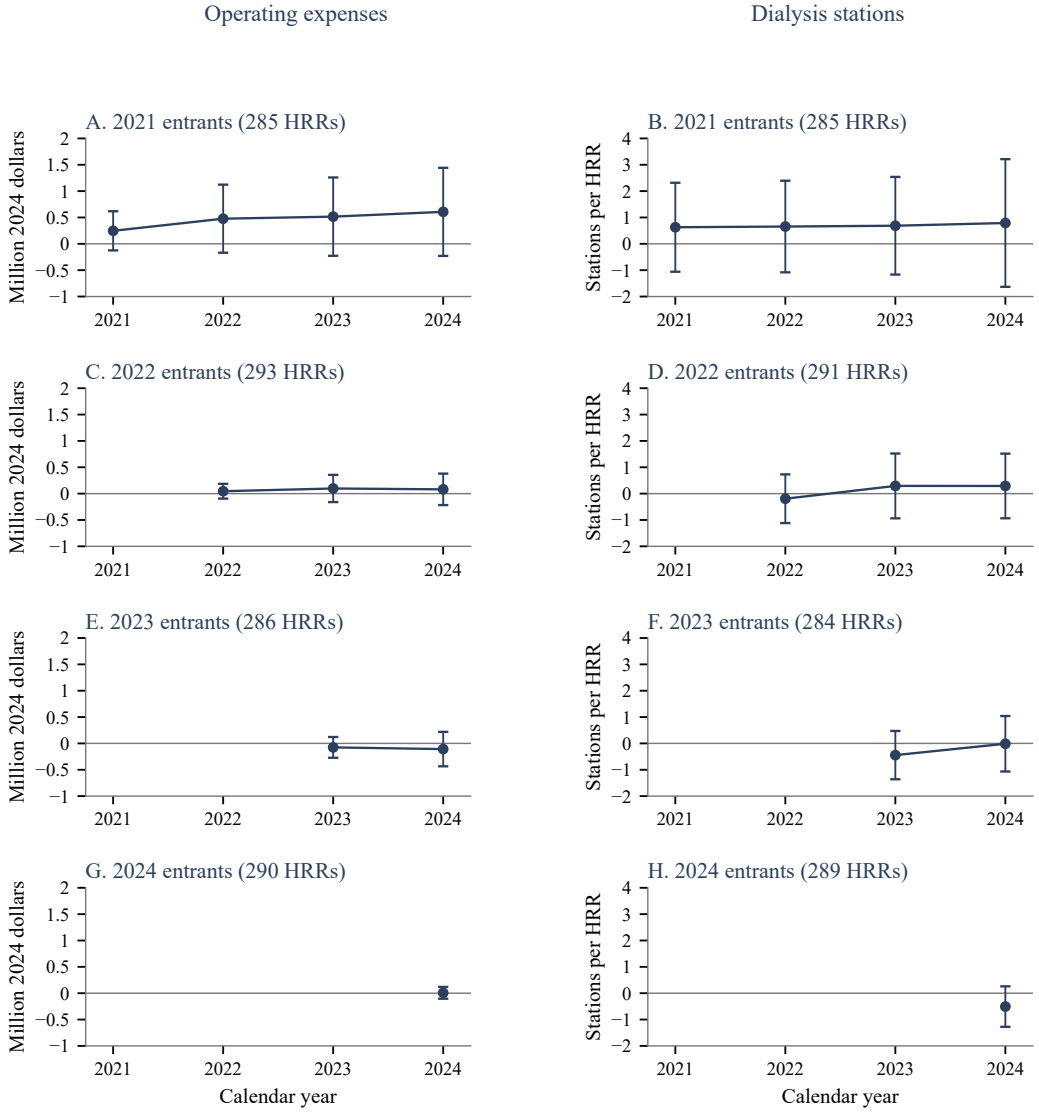
Cohort / year	HRR effect	HRR control mean	Facility effect	Facility control mean	Facilities / HRRs
2022	−46.09	60.31	−58.51	93.51	170 / 93
	(36.82)		(58.13)		
2023	−41.86	63.70	−54.95	100.31	168 / 93
	(38.69)		(62.17)		
2024	−2.75	35.45	0.88	55.83	168 / 93
	(13.70)		(22.44)		
Fixed HRRs	287				
Panel B. 2021 entrants					
2022	25.14*	18.02	25.89	46.30	131 / 71
	(14.04)		(32.02)		
2023	26.14*	23.64	23.60	59.98	132 / 71
	(14.57)		(25.30)		
2024	26.69*	21.25	14.36	53.91	132 / 71
	(14.48)		(25.22)		
Average	—	—	21.28	53.43	395 / 71
			(26.81)		
Fixed HRRs	288				
Panel C. 2022 entrants					
2023	39.21*	12.95	102.55*	41.05	97 / 65
	(22.89)		(57.14)		
2024	19.80	20.92	46.50	66.29	97 / 65
	(19.50)		(53.19)		
Average	—	—	74.52	53.67	194 / 65
			(52.18)		
Fixed HRRs	295				
Panel D. 2023 entrants					
2024	−28.58**	28.81	−139.65***	131.24	66 / 43
	(12.09)		(58.90)		
Average	—	—	−139.65***	131.24	66 / 43
			(58.90)		
Fixed HRRs	292				

Notes: Training counts cover all payers. HRR and facility estimates follow equations (2) and (A1), using the 2019 home-treatment baseline. Qualifying reports last 330–400 days and cover at least 90 percent of the calendar year; cross-year totals are permitted. Follow-up starts after entry. Facility comparisons condition on entry and reporting. The final column counts facility observations and their HRRs; averages count facility-years and weight annual coefficients equally. HRR averages and the 2020 facility average appear in Tables 5 and A6. Parentheses report HRR-clustered HC2 standard errors. Stars denote the 10, 5, and 1 percent levels (*, **, ***) in individual tests using 10,000 region-stratified reassignments under the sharp no-effect null.

D.2 Operating Inputs by Facility Population

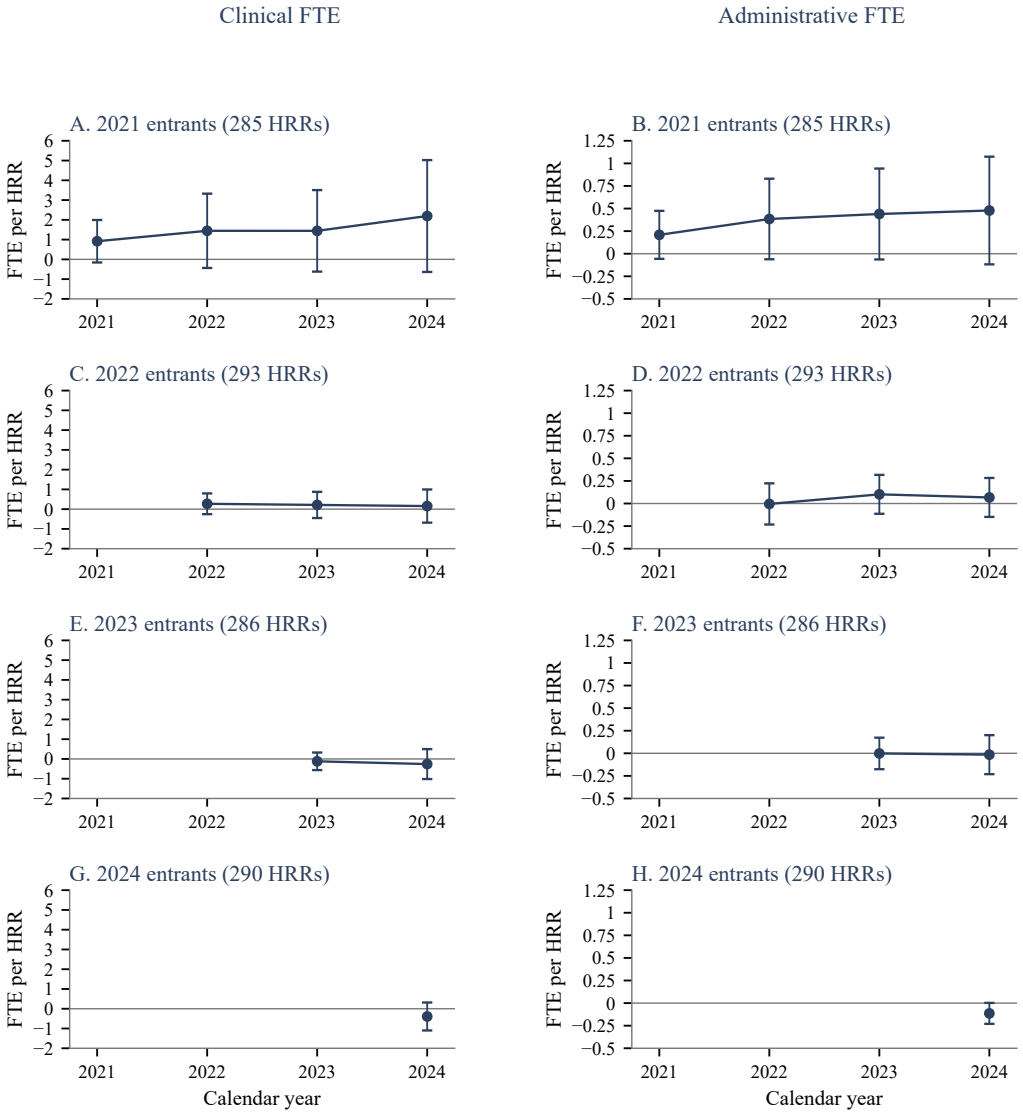
Figures A3, A4, A5, and A6 trace the eight annual entrant input responses summarized in Table 5, from entry through 2024.

Figure A3: Entrant Operating Expenses and Dialysis Capacity



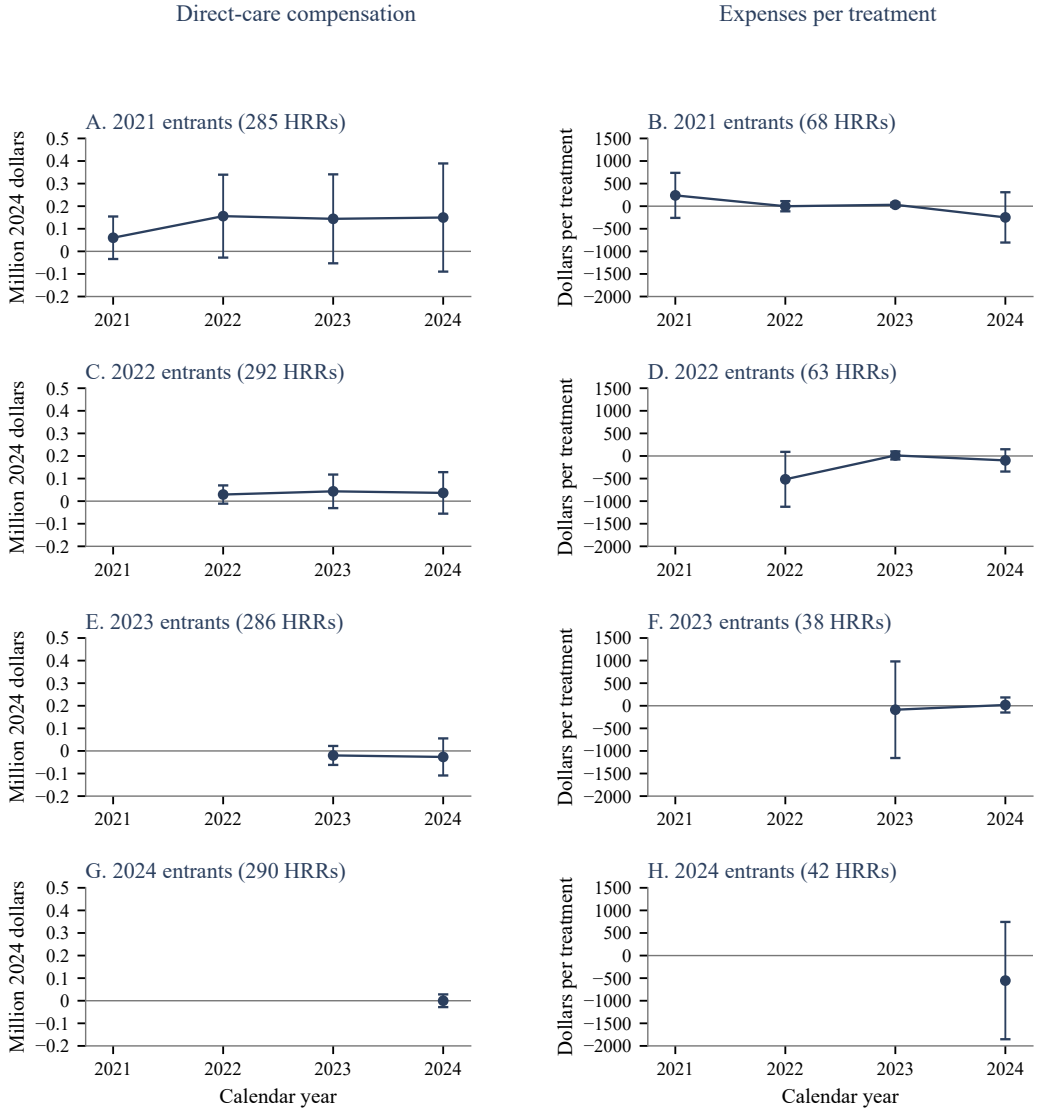
Notes: Annual linear estimates follow equation (2), with corresponding 2019 input baselines and fixed outcome-specific HRRs. Expenses are millions of 2024 dollars per HRR; stations are HRR counts. Entry-year reports may be partial. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Figure A4: Entrant Clinical and Administrative Staffing



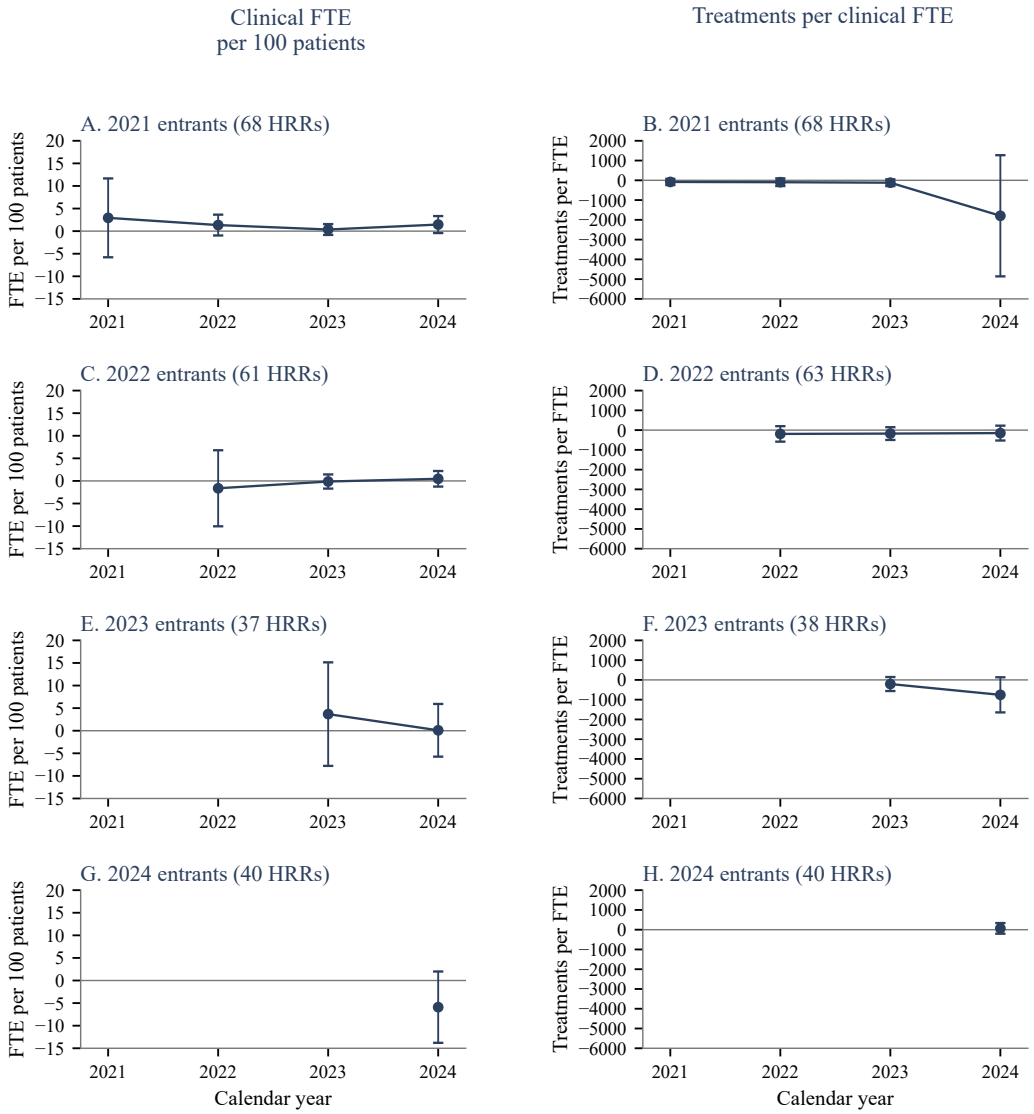
Notes: Annual staffing estimates follow equation (2) on fixed outcome-specific HRRs. Administrative staffing includes management. Follow-up covers entry through 2024; entry-year reports may be partial. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Figure A5: Entrant Compensation and Expenses per Treatment



Notes: Annual estimates follow equation (2) on fixed outcome-specific HRRs. Compensation is millions of 2024 dollars per HRR; expenses per treatment are 2024 dollars, using matched reports and positive all-payer treatment denominators. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Figure A6: Entrant Staffing Intensity and Treatments per Clinical FTE



Notes: Annual estimates follow equation (2) on fixed outcome-specific HRRs. Ratios require matched reports and positive denominators; patient and treatment counts cover all payers. Empty cohorts are excluded. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Table A13 reports the eight average input responses for incumbents and the covered market, together with their 2020 contrasts. All averages are imprecise in both populations. Table A14 gives the annual 2021–2024 entrant estimates underlying the averages in Table 5; the transition-cohort averages appear in Table A6. Each outcome retains a fixed subset of its population's main HRRs and the corresponding 2019 input baseline. Totals include known empty cohorts and verified contributions; matched ratios require positive observed denominators and represent narrower populations. The tables report actual fitted observations and HRRs. These inputs include resources shared across dialysis modalities, limiting their attribution to home-service expansion.

Table A13: Incumbent and Covered-Market Operating Inputs

Outcome	2021–2024 average	2020	Control mean	HRR-years	HRRs
Panel A. Incumbents					
Operating expenses (million dollars)	0.249	0.543	67.409	1,204	301
	(0.872)	(0.530)			
Dialysis stations	9.090	−0.835	371.361	1,204	301
	(6.660)	(2.777)			
Clinical FTEs	0.312	2.592	239.275	1,204	301
	(4.162)	(4.085)			
Administrative FTEs	−1.408	−2.167**	47.827	1,204	301
	(1.230)	(1.272)			
Direct-care compensation (million dollars)	0.139	0.350*	23.041	1,204	301
	(0.290)	(0.211)			
Expenses per treatment (dollars)	−2.27	−5.40*	305.35	1,200	300
	(3.28)	(3.50)			
Clinical FTEs per 100 patients	−0.07	0.22	10.85	1,200	300
	(0.18)	(0.17)			
Treatments per clinical FTE	19.32	15.77	951.65	1,200	300
	(16.18)	(18.36)			
Panel B. Covered market					
Operating expenses (million dollars)	0.424	0.482	70.729	1,204	301
	(0.803)	(0.570)			
Dialysis stations	10.106	−0.180	382.143	1,204	301
	(6.804)	(3.192)			
Clinical FTEs	0.878	2.547	248.727	1,204	301
	(4.185)	(4.129)			
Administrative FTEs	−1.113	−2.233**	50.067	1,204	301
	(1.269)	(1.298)			
Direct-care compensation (million dollars)	0.166	0.339*	23.984	1,204	301
	(0.268)	(0.216)			

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Table A13: Incumbent and Covered-Market Operating Inputs (continued)

Outcome	2021–2024 average	2020	Control mean	HRR-years	HRRs
Expenses per treatment (dollars)	−2.23	−5.45*	308.35	1,200	300
	(3.39)	(3.54)			
Clinical FTEs per 100 patients	−0.02	0.24	10.80	1,200	300
	(0.17)	(0.17)			
Treatments per clinical FTE	13.89	14.67	950.31	1,200	300
	(15.81)	(18.03)			

Notes: Linear estimates follow equation (1). Averages weight 2021–2024 coefficients equally; control means and HRR-years refer to that window. Each outcome retains a fixed observed subset of the 301 main HRRs. Input totals are per HRR; monetary amounts are in 2024 dollars. FTE denotes full-time equivalent. Ratios use positive matched denominators; patients and treatments cover all payers. Parentheses report HRR-clustered HC2 standard errors. Stars denote the 10, 5, and 1 percent levels (*, **, ***) in individual tests using 10,000 region-stratified reassignments under the sharp no-effect null.

Table A14: Annual Operating Inputs of Fixed Entry Cohorts

Outcome / year	2021 entrants	2022 entrants	2023 entrants	2024 entrants
Panel A. Operating expenses (million dollars)				
2021	0.247	—	—	—
	(0.190)	—	—	—
2022	0.477	0.046	—	—
	(0.330)	(0.071)	—	—
2023	0.516	0.097	−0.076	—
	(0.379)	(0.132)	(0.101)	—
2024	0.607	0.081	−0.108	0.007
	(0.427)	(0.152)	(0.167)	(0.057)
Fixed HRRs	285	293	286	290
Panel B. Dialysis stations				
2021	0.629	—	—	—
	(0.862)	—	—	—
2022	0.656	−0.194	—	—
	(0.887)	(0.471)	—	—
2023	0.687	0.292	−0.444	—
	(0.945)	(0.628)	(0.468)	—
2024	0.790	0.291	−0.013	−0.508
	(1.236)	(0.626)	(0.537)	(0.392)
Fixed HRRs	285	291	284	289
Panel C. Clinical FTEs				
2021	0.918	—	—	—
	(0.548)	—	—	—
2022	1.445	0.271	—	—
	(0.959)	(0.268)	—	—
2023	1.443	0.214	−0.119	—
	(1.052)	(0.338)	(0.227)	—

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Table A14: Annual Operating Inputs of Fixed Entry Cohorts (continued)

Outcome / year	2021 entrants	2022 entrants	2023 entrants	2024 entrants
2024	2.194	0.157	−0.259	−0.392
	(1.444)	(0.429)	(0.386)	(0.361)
Fixed HRRs	285	293	286	290
Panel D. Administrative FTEs				
2021	0.209	—	—	—
	(0.136)	—	—	—
2022	0.384	−0.005	—	—
	(0.227)	(0.116)	—	—
2023	0.440*	0.101	−0.002	—
	(0.257)	(0.110)	(0.089)	—
2024	0.478	0.067	−0.015	−0.114*
	(0.304)	(0.110)	(0.110)	(0.059)
Fixed HRRs	285	293	286	290
Panel E. Direct-care compensation (million dollars)				
2021	0.060	—	—	—
	(0.048)	—	—	—
2022	0.156	0.029	—	—
	(0.094)	(0.021)	—	—
2023	0.144	0.043	−0.020	—
	(0.100)	(0.038)	(0.021)	—
2024	0.150	0.037	−0.027	0.000
	(0.122)	(0.047)	(0.042)	(0.014)
Fixed HRRs	285	292	286	290
Panel F. Expenses per treatment (dollars)				
2021	239.11	—	—	—
	(254.36)	—	—	—
2022	−1.06	−516.61*	—	—
	(57.51)	(310.01)	—	—
2023	31.09	12.21	−87.56	—
	(23.61)	(45.42)	(545.57)	—
2024	−248.11	−97.95	17.45	−554.08
	(284.02)	(125.84)	(85.08)	(662.36)
Fixed HRRs	68	63	38	42
Panel G. Clinical FTEs per 100 patients				
2021	2.94	—	—	—
	(4.45)	—	—	—
2022	1.34	−1.63	—	—
	(1.18)	(4.29)	—	—
2023	0.36	−0.13	3.68	—
	(0.61)	(0.80)	(5.85)	—
2024	1.46	0.47	0.09	−5.90
	(0.96)	(0.88)	(2.98)	(4.03)
Fixed HRRs	68	61	37	40

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Table A14: Annual Operating Inputs of Fixed Entry Cohorts (continued)

Outcome / year	2021 entrants	2022 entrants	2023 entrants	2024 entrants
Panel H. Treatments per clinical FTE				
2021	−82.72	—	—	—
	(80.20)	—	—	—
2022	−95.92	−192.84	—	—
	(98.92)	(199.64)	—	—
2023	−120.15	−175.77	−208.74	—
	(87.30)	(164.51)	(180.41)	—
2024	−1,794.82*	−150.56	−755.98*	66.92
	(1,564.14)	(191.63)	(453.17)	(137.70)
Fixed HRRs	68	63	38	40

Notes: Annual linear estimates follow equation (2) on fixed outcome-specific HRRs from entry through 2024; entry-year reports may be partial. Input totals are per HRR; amounts are 2024 dollars. FTE denotes full-time equivalent. Ratios require matched reports and positive denominators; patients and treatments cover all payers. Each annual observation count equals the displayed HRRs. Dashes indicate years before entry. Parentheses report HRR-clustered HC2 standard errors. Stars denote the 10, 5, and 1 percent levels (*, **, ***) in individual tests using 10,000 region-stratified reassignments under the sharp no-effect null.

D.3 Financial, Clinical and Geographic Outcomes

Table A15 consolidates the broader covered-market outcomes using equation (1) and fixed outcome-specific subsets of the main market sample. It retains every outcome in the four families, showing the 2021–2024 average alongside the 2020 coefficient. Patient-service net income falls by $3.83 million per HRR on average, a statistically significant decline, but the negative $3.02 million transition-year difference precedes payment implementation and complicates attribution to those incentives. Long-term catheter use falls by 0.94 percentage points and fistula use rises by 1.04 percentage points, with both changes statistically significant; the vascular-access index summarizes these related improvements. Mortality, hospitalization, readmission, adequacy, geographic availability, and the all-payer home-treatment share have imprecise average responses.[53] Together, these comparisons leave the link between entrant supply, regional access, and patient welfare uncertain.

[53]Financial outcomes reflect accounting allocations; clinical outcomes depend on patient mix. Both span dialysis modalities and available reports. Geographic measures use documented capability nationwide and fixed demand weights partly based on 2020 health prevalence; actual travel, waiting times, and patient choices are unobserved.

Table A15: Supplementary Covered-Market Outcomes

Outcome	2021–2024 average	2020	Control mean	HRR-years	HRRs
Panel A. Financial outcomes					
Medicare payments (million 2024 dollars)	−2.179	−0.250	27.238	1,196	299
	(1.572)	(0.411)			
Payments per treatment (2024 dollars)	0.757	−0.953	352.901	1,196	299
	(3.218)	(1.702)			
Patient-service net income (million 2024 dollars)	−3.834***	−3.024***	13.830	1,196	299
	(1.525)	(1.319)			
Net income per treatment (2024 dollars)	−15.923	−19.120*	64.556	1,196	299
	(13.407)	(12.313)			
Panel B. Clinical indicators					
Standardized mortality ratio	0.002	0.014	0.975	1,200	300
	(0.014)	(0.018)			
Standardized hospitalization ratio	−0.008	−0.001	0.956	1,200	300
	(0.011)	(0.011)			
Standardized readmission ratio	0.001	0.000	0.997	1,196	299
	(0.015)	(0.021)			
Vascular-access index	−0.204***	—	0.920	1,200	300
	(0.071)	—			
Dialysis adequacy fraction	−0.002	0.000	0.968	1,200	300
	(0.002)	(0.001)			
Long-term catheter share (percent)	−0.937***	—	16.822	1,200	300
	(0.354)	—			
Fistula share (percent)	1.043**	0.200	60.671	1,200	300
	(0.435)	(0.303)			
Panel C. Geographic availability					
Nearest documented home-capable facility (miles)	−0.393	−0.485	12.066	1,204	301
	(0.435)	(0.388)			
Documented home-capable facilities within 15 miles	0.030	−0.025	6.302	1,204	301
	(0.116)	(0.056)			
No documented home-capable facility within 30 miles (percent)	0.058	0.025	8.377	1,204	301
	(0.434)	(0.135)			
Panel D. Home-treatment share					
All-payer home treatments as a share of all treatments (percent)	0.547	1.461**	18.225	1,196	299
	(0.543)	(0.787)			

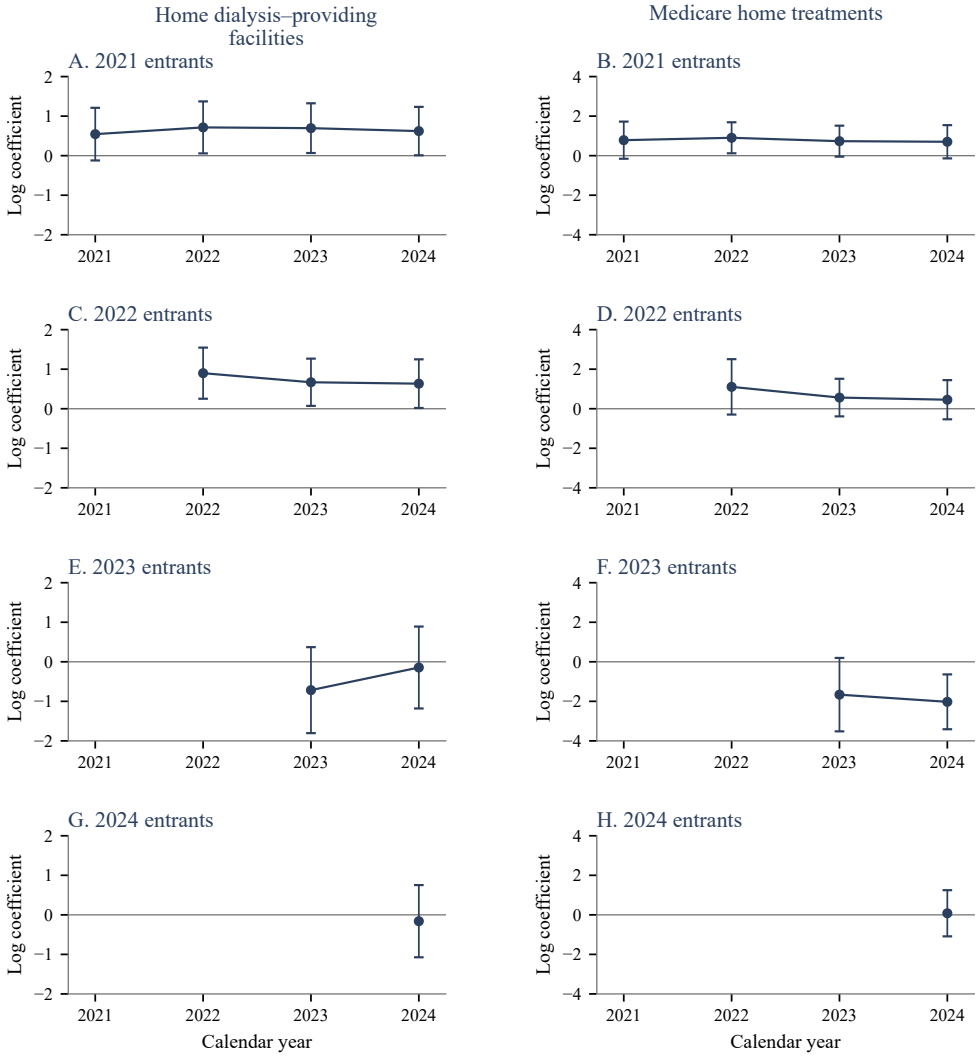
Notes: Estimates follow equation (1) on fixed outcome-specific HRRs. The first column averages 2021–2024 coefficients equally; the second reports 2020. Control means and sample counts refer to 2021–2024. Totals retain verified contributions; ratios require matched reports. Share effects are percentage points except adequacy, a fraction. Geographic measures use fixed demand weights and documented home-service capability. Parentheses report HRR-clustered HC2 standard errors. Stars indicate the 10, 5, and 1 percent levels (*, **, ***), using 10,000 region-stratified efficient-score reassignments under the sharp no-effect null. Dashes denote unavailable observations.

E Robustness Checks

E.1 PPML Estimates of the Main Outcomes

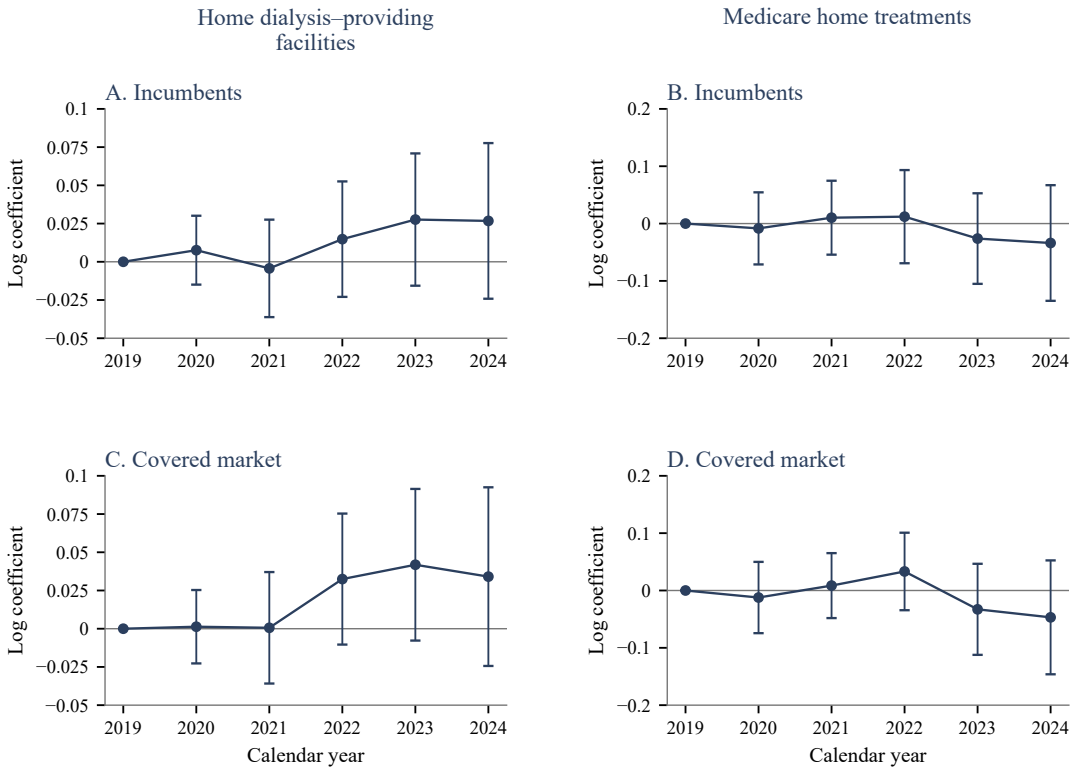
Figures A7 and A8 show the annual PPML supply responses for entry cohorts, incumbents, and the covered market.

Figure A7: Annual PPML responses of fixed entry-cohort home dialysis supply



Notes: PPML estimates follow equation (2) with an exponential link and $\log(1+Y_{h,2019})$ baselines. Coefficients are in log units; separation may reduce fitted support. Table A16 reports annual estimates and averages. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Figure A8: Annual PPML responses of incumbent and covered-market home dialysis supply



Notes: PPML estimates follow equation (1) with an exponential link, relative to 2019. Coefficients are in log units; separation may reduce fitted support. Table A16 reports annual estimates and averages. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Table A16 reports annual log coefficients, equal-weight averages, and fitted support. Cohort baselines enter as $\log(1 + Y_{h,2019})$; the remaining regressor index follows the corresponding primary equation. Separation can reduce the represented observations. Significance follows individual randomization inference; plotted HC2 intervals use a normal approximation.

Table A16: PPML Estimates of Main Supply Outcomes

Year	Providing facilities	Medicare home treatments	Control mean: facilities	Control mean: treatments	HRRs	HRR-years
Panel A. 2020 entrants						
2020	−0.207	0.124	0.188	48.30	290	290
	(0.397)	(0.830)				
2021	−0.143	−0.135	0.252	177.87	290	290
	(0.314)	(0.524)				
2022	−0.244	−0.197	0.267	271.71	290	290
	(0.321)	(0.484)				

Continued on next page

Table A16: PPML Estimates of Main Supply Outcomes (continued)

Year	Providing facilities	Medicare home treatments	Control mean: facilities	Control mean: treatments	HRRs	HRR-years
2023	0.013	−0.568	0.272	339.19	290	290
	(0.303)	(0.568)				
2024	−0.050	−0.579	0.267	363.93	290	290
	(0.336)	(0.585)				
Average	−0.126	−0.271	0.250	240.20	290	1,450
	(0.308)	(0.515)				
Panel B. 2021 entrants						
2021	0.545	0.784	0.156	46.16	290	290
	(0.340)	(0.479)				
2022	0.715*	0.904*	0.180	134.62	290	290
	(0.335)	(0.400)				
2023	0.696*	0.733*	0.185	195.34	290	290
	(0.321)	(0.400)				
2024	0.622*	0.703	0.205	257.63	290	290
	(0.312)	(0.429)				
Average	0.644*	0.781*	0.182	158.44	290	1,160
	(0.321)	(0.413)				
Panel C. 2022 entrants						
2022	0.899**	1.106	0.112	24.15	296	296
	(0.330)	(0.715)				
2023	0.669**	0.564	0.165	93.66	296	296
	(0.305)	(0.485)				
2024	0.634*	0.456	0.165	143.62	296	296
	(0.314)	(0.506)				
Average	0.734**	0.708	0.147	87.14	296	888
	(0.301)	(0.520)				
Panel D. 2023 entrants						
2023	−0.717	−1.662	0.184	51.77	211	211
	(0.555)	(0.947)				
2024	−0.145	−2.025**	0.211	208.38	211	211
	(0.528)	(0.708)				
Average	−0.431	−1.843**	0.197	130.07	211	422
	(0.523)	(0.782)				
Panel E. 2024 entrants						
2024	−0.160	0.080	0.092	32.06	295	295
	(0.466)	(0.596)				
Average	−0.160	0.080	0.092	32.06	295	295
	(0.466)	(0.596)				
Panel F. Incumbents						
2020	0.008	−0.008	10.537	17,737.51	296	296
	(0.011)	(0.032)				
2021	−0.004	0.010	10.639	15,440.19	296	296
	(0.016)	(0.033)				

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Table A16: PPML Estimates of Main Supply Outcomes (continued)

Year	Providing facilities	Medicare home treatments	Control mean: facilities	Control mean: treatments	HRRs	HRR-years
2022	0.015	0.012	10.395	13,947.74	296	296
	(0.019)	(0.041)				
2023	0.028	−0.026	10.122	12,975.88	296	296
	(0.022)	(0.040)				
2024	0.027	−0.034	9.873	12,283.93	296	296
	(0.026)	(0.051)				
Average	0.016	−0.009	10.257	13,661.93	296	1,184
	(0.019)	(0.038)				
Panel G. Covered market						
2020	0.001	−0.012	10.611	17,584.09	299	299
	(0.012)	(0.032)				
2021	0.001	0.009	10.986	15,595.82	299	299
	(0.019)	(0.029)				
2022	0.032	0.033	10.966	14,414.62	299	299
	(0.022)	(0.034)				
2023	0.042*	−0.033	10.913	13,689.70	299	299
	(0.025)	(0.041)				
2024	0.034	−0.047	10.851	13,332.60	299	299
	(0.030)	(0.051)				
Average	0.027	−0.009	10.929	14,258.18	299	1,196
	(0.022)	(0.034)				

Notes: Poisson pseudo-maximum likelihood (PPML) estimates report log coefficients; control means retain the original outcome units. Cohort baselines enter as $\log(1 + Y_{h,2019})$. Cohort averages cover entry through 2024, including 2020 for the transition cohort; incumbent and market averages cover 2021–2024. HRRs and HRR-years report fitted support after separation. Parentheses report HRR-clustered HC2 standard errors. Stars denote the 10, 5, and 1 percent levels (*, **, ***) in individual tests using 10,000 region-stratified reassignments under the sharp no-effect null.

E.2 Influential Markets and Complete Facility Histories

Table A17 compares the main estimates with three sample restrictions: excluding the largest baseline market, excluding the three largest, and requiring resolved provision and treatment quantities for every original facility in every required year. Markets are ranked by their reconciled 2019 eligible-facility counts. The restrictions retain the linear specifications and assess sensitivity to the represented markets, with complete facility histories substantially narrowing incumbent and market support.[54] Complete reporting in the selected sample leaves the process governing missing observations unresolved.

[54]For the supplementary 2020 cohort, this sensitivity comparison averages 2021–2024; its entry-inclusive average in Table A6 covers 2020–2024. All other cohorts are averaged from entry through 2024.

Table A17: Main Supply Estimates under Alternative Sample Restrictions

Population / restriction	Providing facilities	Medicare home treatments	HRRs	HRR-years
Panel A. 2020 entrants				
Main sample	−0.047	−108.73	290	1,160
	(0.062)	(99.00)		
Exclude largest market	−0.053	−118.82	289	1,156
	(0.063)	(100.58)		
Exclude three largest markets	−0.023	−73.53	287	1,148
	(0.058)	(96.17)		
Complete facility histories	−0.001	−102.92	272	1,088
	(0.053)	(94.28)		
Panel B. 2021 entrants				
Main sample	0.125*	183.01*	290	1,160
	(0.073)	(102.86)		
Exclude largest market	0.127*	181.16*	289	1,156
	(0.074)	(102.90)		
Exclude three largest markets	0.117	160.12	287	1,148
	(0.074)	(101.61)		
Complete facility histories	0.106	172.54*	281	1,124
	(0.069)	(97.99)		
Panel C. 2022 entrants				
Main sample	0.134**	86.12	296	888
	(0.063)	(66.15)		
Exclude largest market	0.136**	84.77	295	885
	(0.063)	(66.13)		
Exclude three largest markets	0.133**	72.92	293	879
	(0.061)	(64.86)		
Complete facility histories	0.109*	85.34	288	864
	(0.056)	(66.54)		
Panel D. 2023 entrants				
Main sample	−0.051	−90.74**	293	586
	(0.045)	(33.49)		
Exclude largest market	−0.061	−95.39***	292	584
	(0.045)	(33.55)		
Exclude three largest markets	−0.057	−93.60***	290	580
	(0.045)	(34.56)		
Complete facility histories	−0.068	−91.91***	290	580
	(0.042)	(33.62)		
Panel E. 2024 entrants				
Main sample	−0.015	−2.71	295	295
	(0.039)	(17.23)		
Exclude largest market	−0.012	−1.33	294	294
	(0.039)	(17.23)		
Exclude three largest markets	−0.018	−2.20	292	292
	(0.039)	(17.37)		

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Table A17: Main Supply Estimates under Alternative Sample Restrictions (continued)

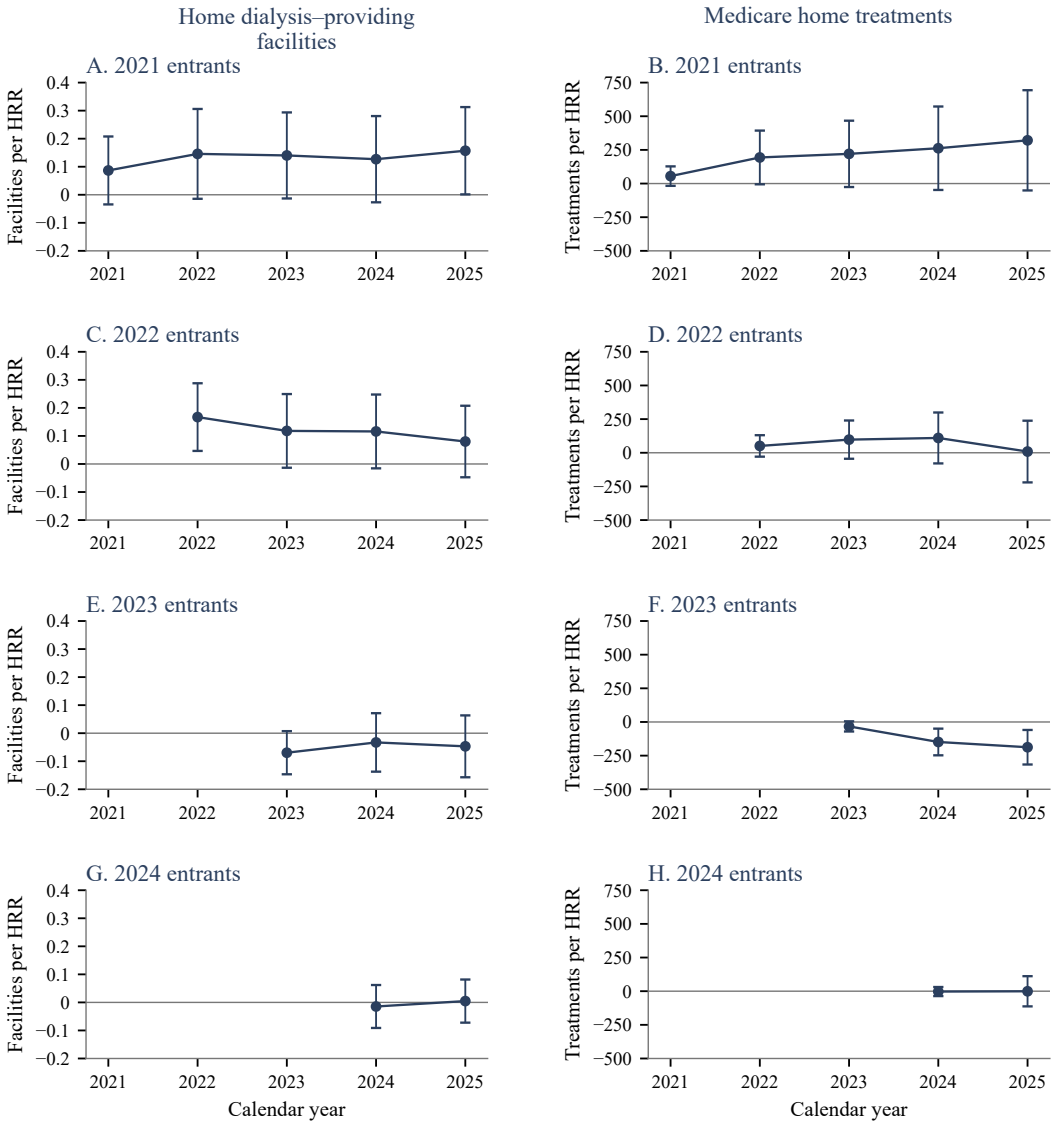
Population / restriction	Providing facilities	Medicare home treatments	HRRs	HRR-years
Complete facility histories	−0.016	−3.14	293	293
	(0.039)	(17.21)		
Panel F. Incumbents				
Main sample	0.228	−135.62	301	1,204
	(0.192)	(640.82)		
Exclude largest market	0.211	−177.84	300	1,200
	(0.191)	(644.60)		
Exclude three largest markets	0.219	−337.38	298	1,192
	(0.192)	(614.94)		
Complete facility histories	0.249	−102.20	97	388
	(0.184)	(648.89)		
Panel G. Covered market				
Main sample	0.299	−73.52	301	1,204
	(0.243)	(579.57)		
Exclude largest market	0.275	−135.26	300	1,200
	(0.243)	(583.00)		
Exclude three largest markets	0.318	−252.13	298	1,192
	(0.242)	(556.30)		
Complete facility histories	0.159	−520.79	86	344
	(0.224)	(755.75)		

Notes: Linear specifications are unchanged. Averages cover 2021–2024 for incumbents, the covered market, and the 2020 cohort, and entry through 2024 for later cohorts. Market size is the 2019 eligible-facility count within each main sample. Complete histories require both outcomes for every original facility and required year, including documented exit zeros. This restriction changes the represented population and does not establish random missingness. Provider counts and Medicare home-treatment equivalents are per HRR. Parentheses report HRR-clustered HC2 standard errors. Stars denote the 10, 5, and 1 percent levels (*, **, ***) in individual tests using 10,000 region-stratified reassignments under the sharp no-effect null.

E.3 Supplementary Follow-Up in 2025

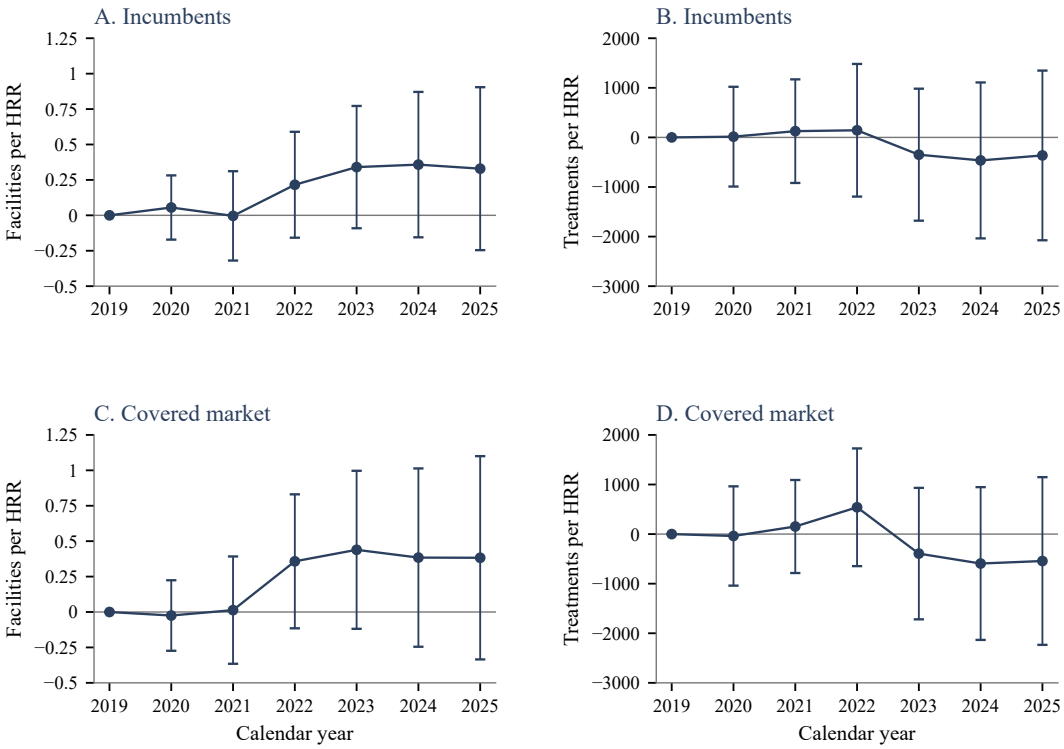
Figures A9 and A10 extend the cohort, incumbent, and covered-market paths through 2025. Available observations are appended before re-estimating the full annual specifications; all earlier observations, main HRR lists, controls, and baseline adjustments are retained. Missing outcomes affect only the additional year. The incumbent and market panels retain 301 HRRs, and the market extension follows the verified 2024 roster. Section 3.2 specifies the remaining data gaps.

Figure A9: Fixed entry-cohort home dialysis supply through 2025



Notes: Figure 2 is extended through 2025 using the original cohort rosters, HRR lists, and pre-2025 observations. Missing 2025 outcomes remain missing. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Figure A10: Incumbent and covered-market home dialysis supply through 2025



Notes: Figure 4 is extended through 2025 on the same 301 HRRs, relative to 2019. The 2025 market follows the verified 2024 roster; unreported facility activity remains unknown. Bars show 95 percent normal-approximation intervals using HRR-clustered HC2 standard errors; significance is assessed separately by randomization inference.

Table A18 reports every annual coefficient, the extended averages, and actual annual support for these three populations. Together with the figures, it assesses continuation of the main supply patterns without allowing incomplete 2025 records to select the pre-2025 sample.

Table A18: Complete Supply Paths through 2025

Year	Providing facilities	Medicare home treatments	Control mean: facilities	Control mean: treatments	HRRs	HRR-years
Panel A. 2021 entrants						
2021	0.087	55.23	0.156	46.16	290	290
	(0.062)	(36.90)				
2022	0.146*	193.86*	0.180	134.62	290	290
	(0.082)	(101.82)				
2023	0.140*	220.57*	0.185	195.34	290	290
	(0.078)	(125.72)				
2024	0.127	262.37	0.205	257.63	290	290
	(0.078)	(158.16)				
2025	0.157*	321.14	0.187	285.51	288	288
	(0.079)	(189.96)				
Average	0.131*	210.63*	0.183	183.65	290	1,448
	(0.074)	(118.80)				
Panel B. 2022 entrants						
2022	0.167***	50.73	0.112	24.15	296	296
	(0.062)	(40.66)				
2023	0.118*	97.60	0.165	93.66	296	296
	(0.067)	(72.52)				
2024	0.116*	110.03	0.165	143.62	296	296
	(0.067)	(96.49)				
2025	0.080	9.01	0.187	244.66	292	292
	(0.065)	(116.93)				
Average	0.120*	66.84	0.157	126.09	296	1,180
	(0.062)	(74.29)				
Panel C. 2023 entrants						
2023	−0.070*	−32.87	0.132	37.12	293	293
	(0.039)	(19.28)				
2024	−0.033	−148.60***	0.151	149.42	293	293
	(0.053)	(50.54)				
2025	−0.047	−187.63***	0.167	209.58	289	289
	(0.056)	(65.46)				
Average	−0.050	−123.04***	0.150	131.79	293	875
	(0.048)	(43.25)				

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Table A18: Complete Supply Paths through 2025 (continued)

Year	Providing facilities	Medicare home treatments	Control mean: facilities	Control mean: treatments	HRRs	HRR-years
Panel D. 2024 entrants						
2024	−0.015	−2.71	0.092	32.06	295	295
	(0.039)	(17.23)				
2025	0.005	−0.70	0.089	103.56	290	290
	(0.039)	(57.21)				
Average	−0.005	−1.71	0.090	67.37	295	585
	(0.038)	(33.87)				
Panel E. Incumbents						
2020	0.055	14.95	10.286	17,315.19	301	301
	(0.116)	(513.20)				
2021	−0.004	126.07	10.386	15,072.57	301	301
	(0.161)	(532.98)				
2022	0.216	143.90	10.148	13,615.65	301	301
	(0.191)	(682.38)				
2023	0.340*	−348.50	9.881	12,666.93	301	301
	(0.220)	(679.08)				
2024	0.358	−463.96	9.638	11,991.46	301	301
	(0.262)	(802.20)				
2025	0.329	−363.45	9.467	11,531.44	301	301
	(0.293)	(872.97)				
Average	0.248	−181.19	9.904	12,975.61	301	1,505
	(0.204)	(670.60)				
Panel F. Covered market						
2020	−0.025	−38.42	10.510	17,416.62	301	301
	(0.127)	(510.93)				
2021	0.013	152.83	10.881	15,447.29	301	301
	(0.193)	(478.58)				
2022	0.358	540.61	10.862	14,277.34	301	301
	(0.241)	(605.60)				
2023	0.439	−393.67	10.810	13,559.32	301	301
	(0.285)	(676.37)				
2024	0.384	−593.87	10.748	13,205.62	301	301
	(0.321)	(785.97)				
2025	0.383	−543.01	10.562	13,057.81	301	301
	(0.366)	(862.79)				
Average	0.315	−167.42	10.772	13,909.48	301	1,505
	(0.258)	(615.05)				

Notes: The regressions append available 2025 outcomes and re-estimate the full annual paths, retaining every pre-2025 observation and the original population-specific HRR lists. Missing 2025 outcomes remain missing. Cohort averages cover entry through 2025; incumbent and market averages cover 2021–2025. The 2025 market follows the verified 2024 roster. Provider counts and Medicare home-treatment equivalents are per HRR. Control means and HRR-years refer to the displayed period; average HRRs count distinct regions across that window. Parentheses report HRR-clustered HC2 standard errors. Stars denote the 10, 5, and 1 percent levels (*, **, ***) in individual tests using 10,000 region-stratified reassignments under the sharp no-effect null.